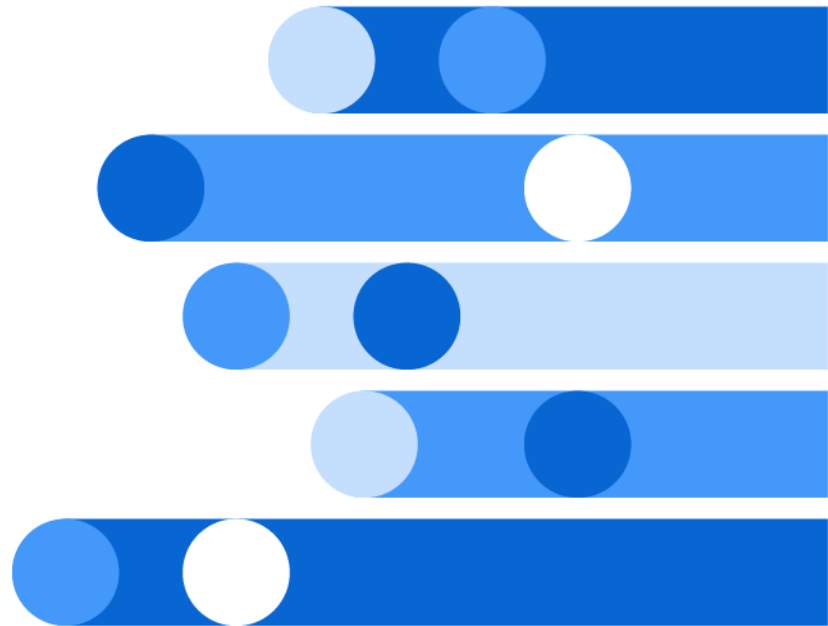




SAS[®] Workflow Manager: User's Guide

2026.05*



* This document might apply to additional versions of the software. Open this document in [SAS Help Center](#) and click on the version in the banner to see all available versions.



The correct bibliographic citation for this manual is as follows: SAS Institute Inc. 2026. *SAS® Workflow Manager: User's Guide*. Cary, NC: SAS Institute Inc.

SAS® Workflow Manager: User's Guide

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May 2026

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Introduction to SAS Workflow Manager

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What Is Business Process Management?

Organizations of all sizes are forced to deal with a changing business environment. The changing marketplace, technological advances, and shifting customer priorities create challenges that businesses must overcome every day. All successful organizations need efficient processes to convert their competencies and resources into value for their customers. Success requires a delicate balance between establishing efficient, repeatable processes, and maintaining the agility to adjust or completely replace these processes to fit current conditions.

Common challenges include the following:

- people acting in concert

- The actions of good managers, along with prior training and experience, largely determine how effectively members of a group can work together to bring about an aggregate result. Yet all of these factors take time to develop, and might never fully develop if the pace of change is high. A well-designed process management system can help by orchestrating—by way of notifications, reminders, delivery of resources, and tracking—the work of many individuals involved in a business process. It can also automate much of the start up and

cleanup in each individual activity. For example, it can automate finding the right forms and locating relevant policies, or automate forwarding to the next person in the process.

interleaving automation

Not all processes can be usefully automated, but even partially automated processes are more efficient than completely manual processes. A well-designed process management system can help identify where automation can have the highest impact, along with an operational framework for deploying and managing automated processes.

performance analysis and optimization

High-level summary results can indicate problems, but detailed analysis is required in order to pinpoint and fix bottlenecks and inefficiencies in operations. A properly implemented process management system can collect detailed metrics on the actual performance of key processes in real time. These metrics give management a concrete basis for making decisions about how and when to make improvements.

Business process management (BPM) is a disciplined approach focused on aligning all aspects of an organization on fulfilling the needs of its clients. It emphasizes integrating technology into the business process such that the process itself drives the business goals, decoupled from the underlying systems and applications. Specifically, BPM emphasizes how the work is done within an organization, in contrast to what a product does.

BPM can also be used to understand relationships between processes and relationships within the organization and across organizational boundaries. The analysis of those relationships, when included in a process model, enable sophisticated, horizontal reporting and analysis.

Critical success factors for BPM include the following:

- understanding the current state business process and client needs
- applying governance and standards based on business policies and practices
- using metric and key performance indicator (KPI) definitions that support measurable business goals

More specifically, a business process is a collection of activities designed to produce a specific output for a particular objective, possibly involving both human and system interactions. Essentially, a process is an ordered sequence of work activities defined with respect to time and place, with a beginning, an end, and clearly defined inputs and outputs: a structure for action.

Initially, BPM focused on the automation of business processes. It has evolved to integrate manual processes in which human interaction takes place in series or parallel with the use of technology. For example, when individual steps in a basic workflow require human intuition or judgment, these steps are assigned to members within the organization. Consequently, the difference between workflow and BPM is not distinct. Generally, workflow management is considered to be a subset of BPM that emphasizes static routing and administration of human tasks. In contrast, a business process might include a combination of automated and manual activities with dynamic routing based on embedded business logic. Today, many products include varying aspects of customization and control, but both

approaches emphasize the elimination of bottlenecks, minimization of redundancies, and improved operational efficiency.

In short, workflow systems can be thought of as a type of operating system for the enterprise. The function of this system is to orchestrate and track work, whether automated or carried out by people. In the same way that databases capture what an organization consumes and produces, workflow systems encapsulate how the organization operates.

Why Use SAS Workflow Manager?

Integration and interoperability of applications and data have improved with the emergence of the web, middleware technologies, enterprise application integration (EAI) efforts, and adherence to software standards like Java, J2EE, and XML. However, businesses in many industries manage the integration of applications, data, and people by adopting business process management (BPM) practices. Popular workflow features include the following:

- systematic routing of tasks requiring manual intervention
- automated triggering of basic actions and alerts

Process management systems offer the option to define, automate, audit, and refine business operations by leveraging the web, middleware, and standards to more efficiently and effectively manage by process. SAS Workflow Manager provides the tools to rapidly integrate fundamental workflow management into business operations and business offerings based on SAS solutions and products.

SAS Workflow Manager on SAS Viya Platform

SAS Workflow Manager integrates with the platform workflow service, which is based on an open-source engine that complies with the Business Process Model and Notation (BPMN) 2.0 standard. It supports industry standards for visual representation and behavior, and it provides support for established workflow patterns. Specifically, SAS Workflow Manager supports workflows that are compliant with the BPMN 2.0 standard. SAS Workflow Manager is efficient and scalable, and it integrates with other standard business process implementations.

The Workflow Definition Lifecycle

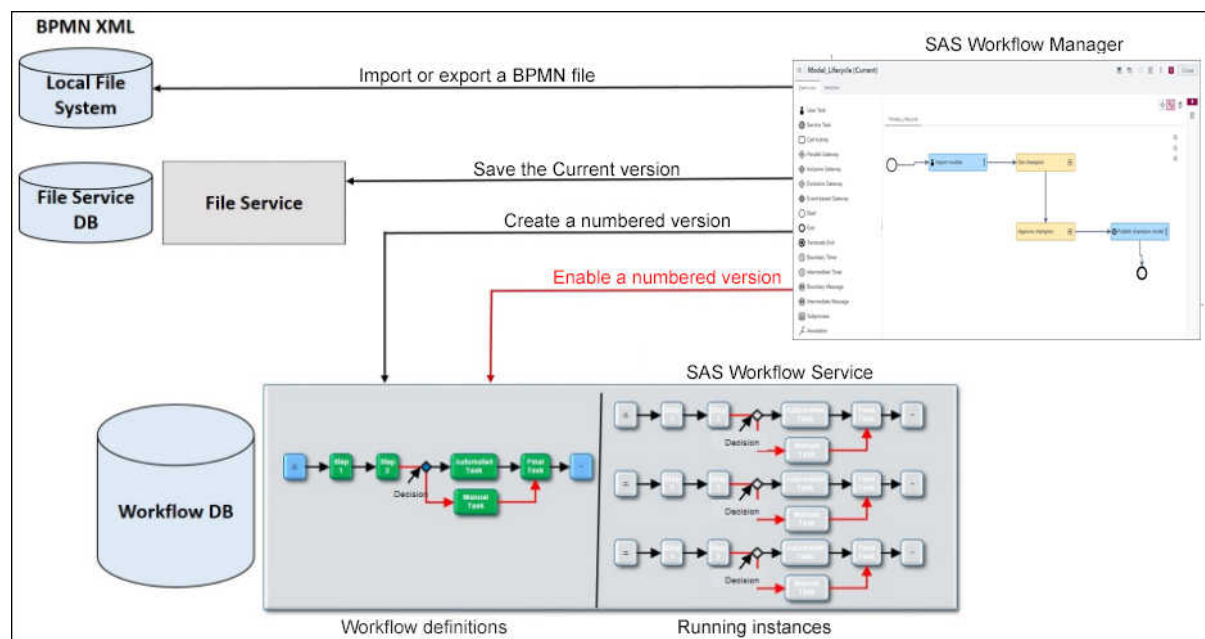
To develop and activate workflow definitions with SAS Workflow Manager:

- 1 Develop a draft of your workflow definition. You can create a new workflow definition or import an existing definition and modify it.
- 2 Save the working draft of your definition. SAS Workflow Manager saves the draft as the Current version. Only you can view and edit your Current version.
- 3 Create a numbered version of your workflow. Any user that has the appropriate permissions can view numbered versions of workflow definitions.
- 4 Activate a specific version of the workflow. In order for end-user client applications to be able to create running instances of the workflow, you must activate a specific version of a workflow definition.

Both the [Quick Start Tutorial](#) and “[Create and Activate a Workflow Definition](#)” on [page 10](#) describe these steps in detail.

Note: Workflow definitions are stored in the BPMN 2.0 XML format.

Figure 1.1 The Workflow Lifecycle



If you determine that you need to make changes to a workflow that has already been activated, you can edit the workflow definition, create a new version, and activate the new version. Client applications can create instances of the new version without affecting any running instances of the old version. Instances of the old version terminate as the workflows are completed.

Only one version of a workflow can be active at a time. As soon as you activate the new version, all new instances of the workflow use the new version of the definition. The older version of the definition is disabled.

Note: Workflow instances are started by the solution. Workflow tasks are managed by the solution while certain administrative activities are available from the **Instances** tab from within SAS Workflow Manager.

Sign In to SAS Workflow Manager

Note: If you are already signed in to a SAS Viya platform app, you can access SAS Workflow Manager by clicking  ⇒ **Manage Workflows**.

To sign in to SAS Workflow Manager:

- 1 In the address bar of your web browser, enter the URL for SAS Workflow Manager and press **Enter**. The Sign In page appears.

Note: Contact your system administrator if you need the URL for SAS Workflow Manager. The default URL is `http://host_name/SASWorkflowManager`.

- 2 Enter a user ID and password.
- 3 Click **Sign In**.

Install SAS Workflow Manager as a Progressive Web App

You can install SAS Workflow Manager as a Progressive Web App (PWA). The benefits of using SAS Workflow Manager as a PWA include the following:

- Application persistence – By default, your session will never time out, so you can restart your work more quickly. Your administrator can enable a time-out for the PWA using a configuration property in SAS Environment Manager. For more information, see [“sas.htmlcommons” in SAS Viya Platform: Configuration Properties](#).

- Performance – When installed as a PWA, SAS Workflow Manager typically starts faster than when it is accessed in the browser.
- Desktop experience – Installing SAS Workflow Manager as a PWA confers all the benefits of a traditional installation. You can launch SAS Workflow Manager from the Start menu or Taskbar, and you do not need to sort through countless tabs to find the correct instances of SAS Workflow Manager.
- Renaming – You can rename each PWA instance to quickly access different environments such as development, test, or production servers.

To install SAS Workflow Manager as a PWA:

Note: SAS Workflow Manager must be deployed with TLS and HTTPS enabled.

- 1 Open SAS Workflow Manager in a Chromium-based web browser and sign in.
- 2 Do one of the following:
 - Click the appropriate icon at the end of the address bar, and then select **Install SAS**.
 - Open the web browser's **More** menu and select **Install SAS**.

Note: The **Install SAS** option might be located under another menu option. For example, within the Microsoft Edge browser it is located under **Apps**.

SAS Workflow Manager is now installed as a desktop program named **SAS**. After one web application has been installed as a PWA, all other SAS applications are accessible in the PWA.

To uninstall the PWA for SAS Workflow Manager:

- 1 Open the PWA for SAS Workflow Manager.
- 2 On the PWA menu, select **Uninstall SAS**.

Changing Node Style in a Workflow Definition

The default node style for SAS Workflow Manager is compact nodes. SAS Workflow Manager also supports the legacy, standard nodes that are larger and allow for resizing.

To change the node style of a workflow definition:

- 1 Double-click the appropriate workflow definition or create a new one.

For more information, see [“Create and Activate a Workflow Definition”](#) on page 10.

- 2 Click  on the workflow definition diagram, next to the zoom icons.
- 3 Select your preferred node style.
- 4 Click  to save the workflow definition.

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About Workflow Definitions

A workflow is a series of tasks, together with the participants and the logic that are required to complete the tasks. A workflow definition is a model of a workflow. A workflow definition can include events, gateways, activities, and other elements. SAS Workflow Manager enables you to create a visual model of the workflow definition as a node-link diagram. When you save the workflow definition, it is saved as a Business Process Model Notation (BPMN) document that conforms to the BPMN 2.0 standard.

Create and Activate a Workflow Definition

TIP Before you model your workflow in SAS Workflow Manager, review the workflow patterns described in [Chapter 11, “Understanding Workflow Patterns,” on page 87](#). Using the right workflow pattern for the right purpose helps ensure that your workflow is efficient and produces the intended results.

- 1 Click  to navigate to the **Definitions** category view.
- 2 Click **New Definition**. The New Definition window appears.
- 3 (Optional) Enter a name for the definition if you do not want to use the default name. Definition names must start with a letter and can contain only alphanumeric characters and underscores (_). Definition names are limited to 255 characters and must be unique.
- 4 (Optional) Enter a description for the new definition.
- 5 Specify a client identifier for the definition. The client identifier is the name of the SAS solution that will create instances of the workflow. For more information, see [“Specify a Client Identifier” on page 14](#).
- 6 Click **OK**. SAS Workflow Manager opens the new definition and displays the **Definition** tab. You define the main process for your workflow on this tab. Any subprocesses that you add to the workflow are displayed on separate tabs.

Note: The default node style is compact nodes. For more information, see [“Changing Node Style in a Workflow Definition” on page 6](#).

TIP Click ↶ to undo your last action.

To select a time interval on which to base the undo function, click and hold ↶.

Automatic save keeps your changes to your working (Current) version of the diagram. Your changes are saved as the Current version and do not affect any changes that were saved as a numbered version. For more information, see [“Manage Versions of Workflow Definitions” on page 25](#).

All actions that were automatically saved within the selected time interval are undone. The available time intervals are static, 2-minute intervals that include all of the time that has elapsed since you clicked  or closed your workflow definition.

TIP Click ↷ to recover your last action.

- 7 To add an element, drag the element from the list of elements onto the workflow diagram. The **Start** and **End** elements are automatically added to the workflow diagram.

Note: A workflow definition must contain a **Start** element and at least one **End** or **Terminate end** element.

- 8 Define the data objects that are required by the workflow. You define data objects in the Definition Properties window. For more information, see [“Create and Manage Data Objects” on page 19](#).
- 9 (Optional) Configure the **Start** element.
 - You can configure the element properties. For more information, see [“Edit Element Properties” on page 13](#).
 - You can configure prompts. For more information, see [“Managing Prompts” on page 36](#).
 - You can configure attributes. For more information, see [“Managing Attributes” on page 38](#).
- 10 Add and configure the elements required in the workflow. For more information, see the following topics:
 - [Chapter 3, “Working with User Tasks,” on page 29](#)
 - [Chapter 4, “Working with Service Tasks,” on page 41](#)
 - [Chapter 5, “Working with Subprocesses and Call Activities,” on page 59](#)
 - [Chapter 6, “Using Timers,” on page 65](#)

- Chapter 7, “Using Signals,” on page 69
- Chapter 8, “Using Messages,” on page 73
- Chapter 9, “Using Gateways,” on page 77
- “Add Annotations and Associations” on page 16

- 11 Add a **Terminate end** element to the workflow definition, or keep the **End** element that was added to the workflow definition during creation.

TIP The End element ends the process defined by the workflow path to which it is added, but it allows any parallel paths and subprocesses to complete. The Terminate end element terminates the process and all of its subprocesses.

- 12 Add sequence flows to connect the elements in the workflow. To connect two elements together, click  to enable link mode, and then drag the cursor from the first element to the next.

TIP To move elements in the diagram, click  to enable move mode. You can switch between link mode () and move mode () as needed.

For more information, see “Add Sequence Flows” on page 14.

- 13 (Optional) Specify conditions for the sequence flows. The conditions on sequence flows are evaluated when a workflow instance executes, and the results determine which paths are taken. For more information, see “Add Sequence Flow Conditions” on page 15.
- 14 (Optional) Add tags to the definition. Tags enable you to categorize related definitions on a user-defined basis. For more information, see “Creating and Managing Tags” on page 22 .
- 15 (Optional) Click  to save the workflow definition. The definition is saved as the Current version.

IMPORTANT An application cannot create instances of the Current version of a definition. In order for an application to use a workflow definition, you must specify a client identifier in the definition properties, create a numbered version, and activate the version.

- 16 Create a numbered version of the definition. For more information, see “Manage Versions of Workflow Definitions” on page 25.
- 17 (Optional) Set permissions on the definition. For more information, see “Set Permissions on a Workflow Definition” on page 17.
- 18 Activate the version. After a numbered version of a definition is activated, the solution that is specified by the client identifier can create new instances of the workflow definition. For more information, see “Activate a Version” on page 26.

IMPORTANT Changes to workflow definitions do not take effect until you activate a revised version of the definition. Instances of a workflow that are already running continue to run with the version of the definition with which they were started. When new workflow instances are started, they use the newly activated version of the definition.

- 19 Click **×** to close the definition.

Edit Definition Properties

Note: If you have not created a numbered version of a workflow definition, you can edit all of a definition's properties. After you create a numbered version of a definition, you can edit only the data objects.

- 1 Click  in the object toolbar, which is located in the top upper right of the window.
The **Definition Properties** window appears.
- 2 Edit the definition properties as needed. For more information, see [“Specify a Client Identifier” on page 14](#) and [“Create and Manage Data Objects” on page 19](#).
- 3 Click **OK**.

Edit Element Properties

Click **▶** to hide the properties pane for the selected element.

All elements, except annotation, have a name and description. These properties are displayed on the **Properties** pane. Click  to display the general properties for the selected task.

Element names are limited to 255 characters and descriptions are limited to 1000 characters.

Specify a Client Identifier

The client identifier is the name of the application that will create instances of the workflow.

- 1 Click the tab for the main process in the workflow. You cannot specify a client identifier if the current tab is a subprocess tab.
- 2 Click  in the object toolbar, which is located in the top upper right of the window.

TIP If you have not created a numbered version of a workflow definition, you can edit all of a definition's properties. After you create a numbered version of a definition, you can edit only the description and data objects.

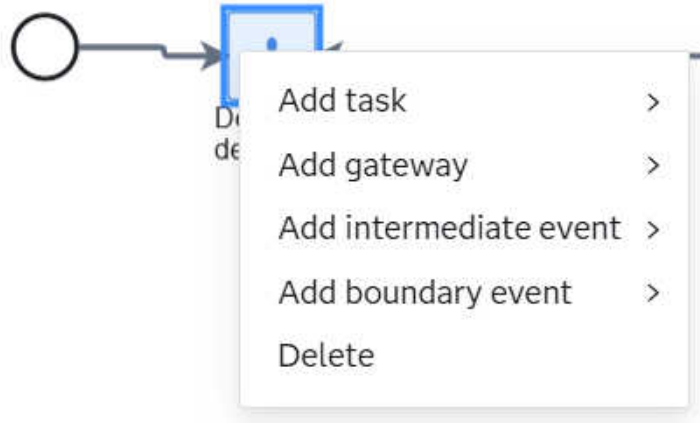
- 3 Select the client identifier and click **OK**.

Add Sequence Flows

A sequence flow connects two events, gateways, or activities in a workflow definition. Sequence flows show the order in which the elements in a workflow are executed. A sequence flow is depicted as a solid arrow.

To connect two elements together, click  to enable link mode, and drag the cursor from the first element to the next.

TIP Click  or right-click on the element to add the next element by using a pop-up menu.



Add Sequence Flow Conditions

You can specify conditions on most outgoing sequence flows. A condition has the following form:

data object operator value-or-data-object

For example:

```
Approved == 'True'
```

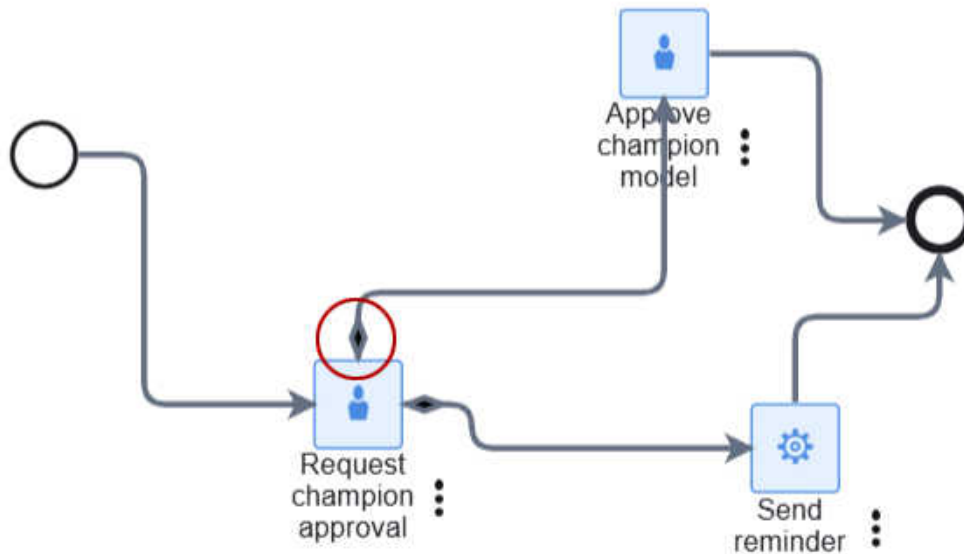
When a workflow instance is run, the condition is evaluated depending on the workflow pattern that the workflow uses. If the condition evaluates to True, the outgoing sequence flow is taken. For more information, see [Chapter 11, “Understanding Workflow Patterns,” on page 87](#).

To add a condition to a sequence flow:

- 1 Select the sequence flow.
- 2 (Optional) Enter a name and description for the sequence flow in the properties pane.
- 3 Click  to view the **Condition** pane.
- 4 Click  to open the Expression Editor window.
- 5 Enter the condition in the expression editor field. You can also use the **Data objects** field and the operator buttons to add object names and operators to the condition.
 - Select a data object, and click ➔ to add the data object to the expression editor field.
 - Select the operator to add the operator to the expression editor field.

TIP In the workflow definition diagram, a sequence flow that contains a condition has a mini-diamond marker on the sequence flow line, as shown in Figure 2.1.

Figure 2.1 Workflow Definition Diagram with Sequence Flow Conditions



Add Annotations and Associations

An annotation is a note that you can associate with events, activities, and gateways. Annotation elements are connected by an association to the executable element that they describe. An association is depicted by a dotted line.

To add an annotation:

- 1 Drag the **Annotation** element onto the diagram.
- 2 On the **Annotation** pane for the annotation, enter the text of the note.
- 3 Click  to enable link mode.
- 4 Drag the cursor from the annotation to the element to which it refers.

Manage Permissions for a Definition

Before an authenticated user can work with workflow definitions and instances in SAS Workflow Manager, they must be given the appropriate permissions.

Any authenticated user can create workflow definitions. Users have full permissions to any definition that they create. Members of the Application Administrators and SAS Administrators groups have full permissions to both workflow definitions and workflow instances. The SAS Workflow Editor, SAS Workflow Viewer, SAS Workflow Definition Administrator, and SAS Workflow Process Administrator groups are no longer defined for you. For more information, see [“Granting Permission to Work with Instances” in SAS Workflow Manager: Administrator’s Guide](#).

Ways to Set Permissions

Permission to work with workflow definitions and instances is controlled in two ways:

- through group membership. For more information, see [“Granting Permission to Work with Instances” in SAS Workflow Manager: Administrator’s Guide](#).
- by setting permissions on specific workflow definitions. For more information, see [“Set Permissions on a Workflow Definition” on page 17](#).

Set Permissions on a Workflow Definition

Note: Permissions that you set on specific workflow definitions override permissions that you set through group membership.

To set permissions on a workflow definition in SAS Workflow Manager:

- 1 In the Definitions view, open the definition for which you want to set permissions.
- 2 Click  in the object toolbar, and select **Authorization**. The Authorization window appears.

IMPORTANT In order to edit the authorization, you must create a new version of the workflow. For more information, see [“Create a Numbered Version” on page 25](#).

- 3 (Optional) Add an identity.
 - a Click . The Select Identity window appears.
 - b Select **User** or **Group** depending on the type of identity that you want to add.
 - c In the search field, enter the identity and choose from the filtered list.
 - d Select the identity that you want to add, and click **OK**.
- 4 (Optional) Delete an identity. To delete an identity, select the identity and click .
- 5 Modify the permissions for the identities. Select or clear the check boxes to set or remove the appropriate permissions.

Permission	Abilities
Read	View the definition.
Update	Modify and delete the definition. Create new versions of the definition.
Enable	Modify, delete, and activate the definition. Create new versions of the definition.
Secure	Modify, delete, and activate the definition. Create new versions of the definition, and set permissions on the definition.

- 6 Click **Save** to save the permissions and close the Authorization window.

Import a Definition

- 1 On the Definitions category page, click **Import**. The Import Definition window appears.
- 2 Click .
- 3 Select the Business Process Model and Notation (BPMN) file that you want to import.
- 4 (Optional) Select a **Client identifier**. The client identifier is the name of the application that creates instances of the workflow.
- 5 Click **OK**.

Export a Definition

You can export workflow definitions as Business Process Model and Notation (BPMN) files.

- 1 Open the definition that you want to export.
- 2 Click  and select **Export**. SAS Workflow Manager adds the BPMN file to the download bar in your browser.

Create and Manage Data Objects

About Data Objects and Scope

Data objects are similar to variables. They hold pieces of business data that are required by the workflow. Global data objects are defined in the main process of a definition and can be used by elements anywhere in the workflow. Local data objects are defined in a subprocess and can be used only within the subprocess and its children.

Data objects in SAS Workflow Manager cannot be null. When you are creating a data object, you can specify a default value for the data object. If you do not specify a default value, the system automatically assigns the following type-based defaults:

Table 2.1 *Type-based Defaults for Data Objects*

Data type	Default value
String	" "
Integer	0
Boolean	false
Decimal	0
DateTime	Current timestamp (for example, 2025-10-06T19:46:09.217Z)

Note: Do not duplicate application data structures in data objects. Instead, define only the data that is referenced in the workflow definition elements. This practice results in a more efficient workflow that is less likely to be affected by changes in the application data model.

Create a Data Object

You can define new data objects by using the Definition Properties window, or you can select the **Create a new data object** option in most fields where you can specify a data object.

To create a data object by using the Definition Properties window:

- 1 Ensure that the correct tab is displayed. To define global objects, click the tab for the main process. To define local data objects, click the tab for the subprocess.
- 2 Click  in the object toolbar, which is located in the top upper right of the window. The Definition Properties window appears.
- 3 Click  above the data objects table. The New Data Object window appears.
- 4 Enter a name for the data object. Data object names must start with a letter and can contain only alphanumeric characters and the underscore (_). Data object names are limited to 50 characters and must be unique within a workflow.
- 5 Select the data type for the data object.
- 6 (Optional) Specify a default value for the data object.
- 7 Click **Save**.
- 8 Click **OK** to close the Definition Properties window.

To specify or create a new data object by using the **Create a new data object** option in a field in which a data object table exists:

- 1 Click , and then select **Data object** in the window that appears.
- 2 In the drop-down menu, select **Create a new data object**. The New Data Object window appears.
- 3 Enter a name for the data object. Data object names must start with a letter and can contain only alphanumeric characters and the underscore (_). Data object names are limited to 50 characters and must be unique within a workflow.
- 4 Select the data type for the data object.
- 5 (Optional) Specify a default value for the data object.

IMPORTANT Though you do not need to specify a default value for the data object, a data object's value cannot be null. You must initialize data objects before they are referenced in any workflow processes. For information about default values for data objects, see [“About Data Objects and Scope”](#) on page 19.

- 6 Click **Save** to close the New Data Object window.
- 7 Click **OK**.

Edit a Data Object

- 1 Ensure that the correct tab is displayed. To edit a global data object, click the tab for the main process. To edit a local data object, click the tab for the subprocess in which the data object is defined.

TIP You can also edit data objects in any field in which a data object table exists by clicking .

- 2 Click  in the application bar above the diagram. The Definition Properties window appears.
- 3 Select the data object, and click . The Edit Data Object window appears.
- 4 Edit the data object properties as needed.
- 5 Click **Save**.
- 6 Click **OK** to close the Definition Properties window.

Delete a Data Object

Note: You cannot delete a data object that is used by an element in the definition.

- 1 Ensure that the correct tab is displayed. To delete global data objects, click the tab for the main process. To delete local data objects, click the tab for the subprocess.

TIP You can also delete data objects in any field where a data object table exists by selecting the data object in the table and clicking .

- 2 Click  in the application bar above the diagram. The Definition Properties window appears.
- 3 Select the data object, and click .

Creating and Managing Tags

About Tags

Tags are alphanumeric labels that are associated with definitions. Tags enable you to categorize related definitions on a user-defined basis. For example, you can define a tag for workflow definitions that is based on the application or organization that owns the definitions.

- Tags are static labels. They have no effect on active workflow instances and cannot be modified at run time.
- You can associate tags with a specific client identifier such as SAS Model Manager or SAS Intelligent Decisioning. Alternatively, you can simply let the tag be associated with any client. Some tags might be private, depending on the client. Private tags can be created or deleted only by the owning client user. For example, some tags associated with the Clinical Workflow identifier, which identifies SAS Clinical Acceleration Repository, are private. These private tags can be created or deleted by only the `sas.clinicalRepository` user.
- Tags do not control access to a definition. They do not prevent the sharing of workflow definitions among different applications. They do not affect whether or how a workflow is active.

Create a Tag

- 1 On the Definitions category page, click  ⇒ **Manage tags**.
The Manage Tags window appears.
- 2 Click .
- 3 Enter the name of the new tag.
- 4 Select the client identifier of the new tag. If you prefer not to specify a client identifier, you can keep the default **All clients** option.
- 5 Click **Save**.

- 6 Click **Close** to close the Manage Tags window.

Add a Tag to a Definition

Note: You must have permission to edit a definition in order to add a tag to it. For more information, see [“Manage Permissions for a Definition” on page 17](#).

- 1 Open the definition to which you want to add tags.
- 2 In the application bar above the diagram, click  ⇒ **Tags**.
The Choose Tags window appears.
If the definition already has a set client identifier, only tags that are associated with that client identifier and tags set to **All clients** are displayed. If the definition does not have a set client identifier, all available tags are displayed.
- 3 Search for a specific tag or filter by the client identifier, and then select the tag or tags that you want to add to the definition.
- 4 Select the tag or tags that you want to add to the definition. Click **OK**.
- 5 Click  to save the definition.

TIP The tags that you add to a definition are listed in the Tags column in the definitions table, unless the definition is the Current version.

Delete a Tag

TIP You cannot delete a tag that has been added to a definition.

- 1 On the Definitions category page, click  ⇒ **Manage tags**.
The Manage Tags window appears.
- 2 Select the tag that you want to delete. Click .
You are prompted to verify the deletion.
- 3 Click **Delete**.
The tag is removed.
- 4 Click **Close** to close the Manage Tags window.

TIP To remove tags from a definition, navigate to the Choose Tags window and deselect the tags that you want to remove from the definition.

Managing Templates

About Templates

You can create custom templates for service tasks, user tasks, and subprocesses to use in other workflow definitions.

For more information about creating service task templates, see [“Create and Use a Service Task Template” on page 55](#).

For more information about creating user task templates, see [“Create and Use a User Task Template” on page 39](#).

For more information about creating a subprocess template, see [“Create and Use a Subprocess Template” on page 62](#).

Delete a Template

- 1 On the Definitions category page, click  ⇒ **Manage templates**.

The Manage Templates window appears.

- 2 Select one or more templates that you want to delete.

TIP To reduce the number of templates that are displayed in the list, click . Then, select which types of templates or client IDs to display.

You can also search for a specific template in the search box.

- 3 Click .

You are prompted to verify the deletion.

- 4 Click **Delete**.

The template or templates are removed.

- 5 Click **Close** to close the Manage Templates window.

Manage Versions of Workflow Definitions

About Versions

Changes to a workflow definition do not take effect until you activate a revised version of the definition. Instances of a workflow that are already running continue to run with the version of the definition with which they were started. When new workflow instances are started, they use the newly activated version of the definition.

The version of a definition that you are in the process of working on is the **Current** version. Only you can see your **Current** version. Other users cannot see changes that you have made to a definition until you create a numbered version of the definition.

Set the Displayed Version

The displayed version is the version whose information is displayed on the **Definition** tab. On the **Versions** tab, a ✓ indicates the displayed version. To change the displayed version, select the version that you want to view and select **Set version**.

Create a Numbered Version

Note: In order to create a numbered version of the definition, you must specify a **Client identifier** in the Definition Properties window.

Note: You cannot save changes to a numbered version. If you edit a numbered version and click , SAS Workflow Manager your changes are saved as the Current version. If a Current version does not exist, SAS Workflow Manager creates it. If a Current version already exists, it is replaced.

To create a new version:

- 1 Click  to create a static numbered version of the workflow definition. The New Version window appears.
- 2 (Optional) Enter any notes that you want to associate with the new version.

TIP Notes can help you identify what has changed from one version to the next.

- 3 Click **Yes**. SAS Workflow Manager assigns the number 1.0 to the version.

Activate a Version

In order for the workflow definition to run a workflow process, you must activate the appropriate version of the workflow definition. Only one version of a workflow definition can be activated at a time. If a version is already active and you activate a different version, the previously active version is disabled and cannot be used by other applications.

IMPORTANT Changes to workflow definitions are not implemented until you activate the revised version of the definition. Instances of a workflow that are already running continue to run with the version of the definition with which they were started. When new workflow instances are started, they use the newly active version of the definition.

Note: You cannot activate the Current version of a definition. To activate a definition, you must first create a numbered version. For more information, see [“Manage Versions of Workflow Definitions” on page 25](#).

To activate a numbered version of a definition, open the version, and click . The version is marked **Active** in the **Status** column.

It is not possible to revert an activation of a workflow definition's version in SAS Workflow Manager. After you have activated any version of a definition, you cannot delete that definition.

Delete Definitions

.....
Note: You cannot delete a definition after any version of it has been activated.
.....

- 1 Click  to open the **Definitions** category view.
- 2 Select the check box for the definition that you want to delete, click , and then select **Delete**. A confirmation message is displayed.
- 3 Click **Yes** in the confirmation to delete the workflow definition, or click **No** to exit the message.

Working with User Tasks

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About User Tasks

User tasks are tasks that are performed by a human participant. Typically, the user tasks in a running workflow are managed through a software application that provides the ability to claim and complete the tasks. For example, SAS Model Manager provides a Tasks category through which users can claim and complete the tasks to which they are assigned. When workflow execution reaches a user task, an instance of the task is created and a potential owner who has access to the task can complete it. The user task then appears in each potential owner's task list.

Add a User Task

- 1 Drag the **User task** object from the list of elements onto the workflow diagram.
 - 2 (Optional) On the **Properties** pane, edit the name and description. For more information, see [“Edit Element Properties” on page 13](#).
 - 3 (Optional) Select **Set a due date** and specify a due date.
 - To enter the due date that is relative to when the task starts, select **When the user task is reached**. If you do not enter a delay, the task becomes due as soon as the task starts. You can select a delay under **Add a delay**.
 - To use a data object to specify the due date, select **Specified in a data object**, and select the data object. The data object type must be datetime. The data object is evaluated when the user task starts and does not change even if the value of the data object changes.
- TIP** The due date is a task property that is available for use by the application that starts the workflow. The due date does not affect the execution of the workflow. To set deadlines that affect the execution of the workflow, use a timer. For more information, see [Chapter 6, “Using Timers,” on page 65](#).
- 4 Add the participants for the task. For more information, see [“Managing Participants” on page 30](#).
 - 5 (Optional) Add any prompts that are needed for the task. For more information, see [“Managing Prompts” on page 36](#).
 - 6 (Optional) Add any attributes that are needed for the task. For more information, see [“Managing Attributes” on page 38](#).

Managing Participants

About Participants and Roles

When you define a user task, you associate users with the task by adding them to the task as participants. A participant can be an individual user or a group of users.

You can assign a participant to one of two roles:

Potential owner

A potential owner can claim and complete a task.

Typically, a user task has a set of potential owners, but only one person can claim and complete the task. You can specify one or more users or groups as potential users.

Excluded Owner

An excluded owner cannot claim or view a task. Excluded owners are typically used when a task needs to be reviewed or approved by at least two people.

Using excluded owners ensures that the same person does not review the task twice and can increase the transparency of an organization's processes.

Note: SAS Workflow Manager supports an additional role: Actual Owner. A participant becomes an actual owner after they claim a task in a running workflow, but you cannot assign this role in a workflow definition.

Participants must be available through the SAS Viya platform Identities service and have the appropriate permissions. For additional information, see [SAS Workflow Manager: Administrator's Guide](#) and [SAS Viya Platform: Identity Management](#).

Note: You can also assign participants by using a data object.

Add Participants to a User Task

- 1 Click  to display the **Participants** properties pane.
- 2 From the **List type** drop-down list, select how the participants are specified:

Specified in a data object

specifies that the list of participants is read from a data object.

Specified in a table

specifies that the list of participants is specified directly on the **Participants** pane.

TIP If the set of potential owners is unknown at design time, you should use a data object to configure participants dynamically. For more information, see [“Specifying Groups as Participants” on page 35](#).

To specify participants by using a data object, select **Listed in this data object** to use an existing data object, or select **Create a new data object**. For more information, see [“Data Object Format for Participants” on page 34](#).

To specify participants by selecting identities, select **Listed in this table**, and complete the following steps for each participant:

- 1 Select the workflow role for the participant. For more information, see [“About Participants and Roles” on page 30](#).
- 2 Click  next to the chosen role and add an owner user or group. The Choose Members window appears.
- 3 In the search field, enter the **User name** or **Group name**.
- 4 Select **User name** or **Group name**, and click **OK** to close the Choose Members window.

TIP If you know the user ID of the user or group that you would like to assign as a potential owner, you can enter it directly into the **Selected members** field.

Note: You can add more than one **User name** or **Group name** at a time.

To remove a participant, click **×** next to the participant's name.

Enable Multiple Participants to Complete a Task

If a user task has multiple potential owners and more than one person must claim and complete the task, you can specify that the task is a multi-instance task. You can use multi-instance tasks to repeat activities in a workflow without defining the same task multiple times.

Note: When you specify a group as a potential owner, the number of potential owners that are assigned to a task and the number of instances that are created depend on how you specify the group. For more information, see [“Specifying Groups as Participants” on page 35](#).

If only one potential owner is needed to complete the task, then follow these steps:

- 1 Click  to access the **Instances** pane of a user task.
- 2 From the **Instance creation type** drop-down list, select **Single instance**.
- 3 (Optional) You can configure the single instance to set or exclude an owner based on a data object or on a transient variable. To configure these settings:
 - a Select **Set actual owner value** or **Set excluded owner value** as the participant role.
 - b Click .
 - c Do one of the following:
 - Select **Data object**, and choose a data object from the drop-down menu to set the actual owner based on the data object.

- Select **Transient variable** and enter the variable information in the text box.

If more than one potential owner must complete the task, the workflow service must create a separate instance of the task for each person. To enable more than one potential owner to complete the task, configure these settings:

- 1 Click  to access the **Instances** pane of a user task.
- 2 From the **Instance creation type** drop-down list, select **New instance for each participant**.
- 3 From the **Run instance** drop-down list, select **Instances run in parallel** or **Instances run sequentially**.

Select **Instances run in parallel** to allow multiple people to work on the task at the same time. Select **Instances run sequentially** to allow only one person to work on the task at a time. Choose the option that is appropriate for your task.

- 4 (Optional) Specify a completion condition. This condition is an expression that controls how multiple instances are handled or how the completion of the user task is determined. The following table shows the criteria for setting an appropriate condition.

For more information, see [“Add a Condition and Instance Variables” on page 33](#).

Table 3.1 Criteria for Completing Multi-instance User Tasks

Create a new instance for each participant is selected	Completion condition is specified	Criteria for Determining When a Task Is Complete
Yes	Yes	The completion condition evaluates to True, and the required number of potential owners claim and complete the task.
Yes	No	Every potential owner claims and completes the task.
No		One of the potential owners claims and completes the task.

Add a Condition and Instance Variables

For multi-instance user tasks, you can specify a condition that indicates when the task is complete.

- 1 Select  to open the Expression Editor window.
- 2 Select an instance variable, and then click  to add the variable to the expression editor field.
- 3 Select the operator to add it to the expression editor field.
- 4 For the second operand in the expression, either enter a literal value or add a second instance variable to the expression editor field.

SAS Workflow Manager provides three predefined variables that you can use to construct a completion condition:

nrOfInstances

specifies the total number of active and completed instances for the task. For example, if you specify a completion condition of `nrOfCompletedInstances>=nrOfInstances/2`, the task is considered complete when half of the potential owners have claimed and completed the task.

nrOfActiveInstances

specifies the total number of instances that can be active at the same time. If the instances are run sequentially, this number must be 1. If the instances are run in parallel, this number cannot exceed the value of **nrOfInstances** (if it is specified).

For example, suppose you have a user task that has ten potential owners and the instances for each user are run in parallel. You can limit the number of people that can claim the task at the same time to two by specifying the following condition:

```
nrOfActiveInstances==2
```

As soon as one of the two participants completes the task, a third participant can claim and complete the next instance of the task.

nrOfCompletedInstances

specifies the number of instances that must be completed. This number cannot exceed the value of **nrOfInstances** (if it is specified).

For example, suppose you have a user task that has five potential owners, but only three people need to complete the task. You would specify the following condition:

```
nrOfCompletedInstances==3
```

If a fourth person claims a parallel task, but does not complete the task before the third person completes the task, the fourth instance of the task is terminated.

Data Object Format for Participants

To specify participants by using a data object, the value of the data object must follow this format:

```
[{ "principalType" : " type", "role" : "role", "principal" : "identity" }]
```

```
<,{ "principalType" : " type", "role" : "role", "principal" : "identity" }...>]
```

type

specify **User** for individual users and **Group** for groups.

TIP If more than one member of a group needs to complete a multi-instance task, the data object must list the name of the group once for each member that must complete the task. For more information, see [“Specifying Groups as Participants” on page 35](#).

role

specify **PotentialOwner** or **ExcludedOwner**. See [“About Participants and Roles” on page 30](#) for descriptions of these roles.

identity

specify an identity that is available through the SAS Viya platform Identities service. See [SAS Viya Platform: Identity Management](#) for additional information.

Note: The data object is evaluated when the user task starts and the user assigned to the task does not change, even if the value of the data object changes.

Specifying Groups as Participants

If you specify a group name only once, then only one member of the group can claim and complete the task. The first member to claim the task becomes the actual owner and can complete the task. The remaining members of the group can no longer claim the task.

If more than one member of a group needs to complete a task, then you must specify the potential owners by using a data object with a multi-instance task. The data object must list the name of the group once for each member that must complete the task. For example, if the group *Approvers* defines ten users, and three users must claim and complete the task, then the data object must contain the following value:

```
[{ "principalType" : "Group", "role" : "PotentialOwner", "principal" : "Approvers" },
{ "principalType" : "Group", "role" : "PotentialOwner", "principal" : "Approvers" },
{ "principalType" : "Group", "role" : "PotentialOwner", "principal" : "Approvers" }]
```

Managing Prompts

About Prompts, Scope, and Reuse

Prompts enable you to request information from workflow participants and store the user's response in data objects. By using prompts, you can set values that are required to evaluate sequence flow expressions or to execute other tasks.

You can define prompts for intermediate messages, start elements, and user tasks. You must define prompts for required values before the values are needed. For example, if a data object value needs to be initialized when a workflow instance is created, define a prompt on the start element. If a value is required by the sequence flow expressions for a gateway, define the prompt on a start element or on a task that is earlier in the workflow than the gateway.

A prompt has the same scope as the data object that is used to store the user's response. Depending on the scope of the data object, a prompt that is defined for one element can be reused by (added to) other user tasks. If the scope of the data object is global, the prompt can be reused anywhere within the workflow definition. If the scope is local, the prompt can be reused only within the process where it is defined and within any children of that process. For more information, see [“About Data Objects and Scope” on page 19](#).

Define a New Prompt

- 1 In the workflow definition diagram, select the **Intermediate message**, **Start** element, or the **User task** for which you want to define a prompt.
- 2 Click  to display the **Prompts** property pane.
- 3 Click   **Add new prompt**. The New Prompt window appears.
- 4 In the **Prompt label** field, enter the text that you want to be displayed for the prompt.
- 5 (Optional) Select **This value is required** if the user is required to enter or select a value for this prompt.
- 6 Select or create the data object in which you want to store the user's response. For more information, see [“Create a Data Object” on page 20](#).
- 7 (Optional) Select the type of attribute to create. Perform one of the following:
 - Enable the user to associate custom prompt attributes with prompts in the workflow definition:

- 1 Select **Use custom prompt for client identifier via attribute**.
 - 2 Select a custom prompt attribute from the **Attributes** drop-down menu.
 - Enable the user to set prompt properties in the workflow definition:
 - 1 Select **Set prompt properties via attribute**.
 - 2 Click  under **Properties**.
The Edit prompt properties window appears.
 - 3 Click .
 - SAS Workflow Manager adds a new row to the **Key-Value** table.
 - 4 In the **Key-Value** table, enter a name and value. Key values must begin with a letter and can contain only alphanumeric characters and underscores. Values can contain only alphanumeric characters and underscores.
- TIP** To delete a key-value pair, select the value in the table and click .
- 5 Enter additional key-value pairs until you have created all that you need.
 - 6 Click **OK** to close the Edit prompt properties window.
 - 8 (Optional) Specify the list of valid display values for the response. To add a new display value complete the following:
 - a Click . SAS Workflow Manager adds a new row to the **Displayed values** table.
 - b In the **Displayed values** table, enter a name and value.
- TIP** To delete a prompt value, select the value in the table and click .
- 9 Click **OK** to close the New Prompt window.

Reuse an Existing Prompt

- 1 Select the **Start** element or **User task** for which you want to define a prompt.
- 2 Click **+ ⇒ Add existing prompts**. The Add Prompts window appears.
- 3 Select the prompts that you want to add. Click **OK**.

TIP If the prompt defines attributes, they are also added. You can view the imported attributes in the **Attributes** properties pane.

Edit a Prompt

- 1 Click  to display the **Prompts** property pane.
- 2 Select a prompt to edit. Click .
The Edit Prompt window appears.
- 3 Edit the prompt as needed.
- 4 Click **OK**.

TIP Your changes are reflected in every user task that uses the prompt.

Delete or Edit a Prompt

- 1 Click  to display the **Prompts** property pane.
- 2 Click  to edit a prompt.
- 3 Click  to delete a prompt.

Note: You can add prompt values to or remove them from an existing prompt definition without affecting previous associations.

Managing Attributes

About Attributes

Attributes are used to associate key-value pairs to user tasks. In some solutions, they are used to enhance the end-user experience. For example, attributes enable a

SAS Model Manager user to choose from a list of models instead of filling in a blank field.

Add Attributes to a User Task

You can add the following attribute values to a user task:

- select_potential_owner
 - show_model_list
- 1 Click  to display the **Attributes** properties pane.
 - 2 Click  to display the Add an Attribute window.
 - 3 Enter a name and value for the attribute. Attribute names are limited to 100 characters.
 - 4 Click **OK**.

Remove an Attribute

- 1 Click  to display the **Attributes** properties pane.
- 2 Click the attribute that you want to remove.
- 3 Click  to remove the attribute from the user task.

Create and Use a User Task Template

The template action enables you to specify a full or partial user task definition and save it as a custom user task template. After you create a user task template, you can use it in other workflow definitions.

Create a Template

- 1 Follow the steps in [“Add a User Task” on page 30](#) to create a user task.
- 2 Click  in the object toolbar. Click **Save As**.
The Save as Template window appears.
- 3 Specify a name for the user task template.
- 4 (Optional) Enter a description for the user task template.

- 5 Select the associated application in the drop-down menu.

Note: The associated application that you select should be the same as the Client Identifier for the workflow definition.

- 6 (Optional) Specify additional data objects that you want to save in the template. The data objects that are used in the user task are already included. For more information, see [“Create and Manage Data Objects” on page 19](#).
- 7 Click **OK** to save the user task template.

TIP You can overwrite the saved template. Make the appropriate changes, click **Save As**, and save it with the same name.

Use a User Task Template In a Workflow Definition

- 1 Drag the **Templated user task** element from the list of elements onto the workflow diagram. The Choose Template window appears.
- 2 Select a template. Perform one of the following actions:
 - In the filter field, start entering a template name, and then select one in the filtered list. Click **Add**.
 - Select a template in the list of all available templates and click **Add**.

All of the fields in the **Actions** pane are populated with the information that is stored in the template.

- 3 (Optional) On the **Properties** pane, edit any of the properties (for example, name and description). Changing the values for properties in this definition does not change the template or other definitions where the template is used. For more information, see [“Edit Element Properties” on page 13](#).
- 4 (Optional) Click  to save your workflow definition.

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About Service Tasks

A service task is a task that invokes an executable business action. These actions are triggered automatically when workflow execution reaches the service task. For example, you can use service tasks to perform the following actions:

- send email to process participants and groups about deadlines or status changes
- communicate with other applications by executing web services
- execute business logic or rules
- integrate the system with other back-end systems

When you define a service task, you select one of six different actions:

Invoke REST web service

invokes a REST web service

Send email

sends email using the SAS Viya platform Mail Service

Send notification

enables you to send separate notifications to individual users or groups using the SAS Viya platform Notification Service

Send group notification

enables you to send a single notification to individual users and groups, and to copy additional users and groups using the SAS Viya platform Notification Service

Invoke Job Execution service

enables you to execute jobs that are defined in the Job Execution service

About Service Task Transient Variables

Transient variables can be used as process variables to pass data. Use transient variables only to store values that are required for the next user task, timer, or message in the workflow. For example, you can use a transient variable when a web service response contains a value that is required as input to another web service call.

Transient variables are identified by an underscore prefix.

In addition to user-defined transient variables, SAS Workflow Manager automatically creates transient variables as part of service task execution. Here are some of the transient variables that are created automatically:

_SAS_ResponseBody

contains the response body.

_SAS_ResponseHeaders

contains the response headers.

_SAS_ServiceTaskID

contains the service task identifier.

_SAS_ResponseStatusCode

contains the response status code of the HTTP call.

Note: Use transient variables to pass data only from within a workflow that is not part of the solution state.

Add a Service Task

- 1 Drag the **Service task** element from the list of elements onto the workflow diagram.
- 2 (Optional) On the **Properties** pane, edit the name and description. For more information, see [“Edit Element Properties” on page 13](#).
- 3 Click  to display the **Action** property pane.
- 4 Select an action.
- 5 Enter the information for the service task. See the following topics:
 - [“The Invoke REST Web Service Action” on page 43](#)
 - [“The Send Email Action” on page 47](#)
 - [“The Send Notification Action” on page 49](#)
 - [“The Send Group Notification Action” on page 51](#)
 - [“The Invoke Job Execution Service Action” on page 52](#)
 - [“Create and Use a Service Task Template” on page 55](#)

For information about the SAS Viya platform APIs, see developer.sas.com.

The Invoke REST Web Service Action

This action enables you to invoke a REST web service over HTTP.

Table 4.1 Properties for the Invoke REST Web Service Action

Field	Description
URL	Specifies the URL of the web service endpoint for the task.
Method	Specifies the HTTP method for the web service such as GET , POST , PUT , and so on.

Field	Description
HTTP request headers	Specifies the names and values of the HTTP header fields in the request. Click  to add a new header. To remove or edit a header, select the header and click either  or  .
Expected result codes	Specifies a list of the HTTP response status code or codes that indicate whether a specific request was successfully completed.
Input type	Specifies the type of the input body. The type can be JSON, XML, or empty (no input body).
Input body	Specifies the content of the input body of the request.
HTTP response headers	Specifies the data object mapping to field values that are part of the HTTP header in the response. Click  to add a new header. To remove or edit a header, select the header and click either  or  .
Output type	Specifies the output type of the response. The type can be either JSON or XML.
Output data objects	Associates data objects with result fields in the response. See “Manage Output Data Objects for REST Web Service Tasks” on page 45 for instructions.

IMPORTANT A user account must be configured for a workflow client in order to make service task calls within a definition. This account must be an LDAP account and must be configured by the solution at start-up, or it must be configured by the system administrator before the creation of any instances. For more information, see [“Configure the Workflow Client” in SAS Workflow Manager: Administrator’s Guide](#).

Note: External requests are not supported at this time. Only relative URLs are supported for the URL field. To prevent unauthorized platforms actions, the following URIs are restricted:

- /workflow
- /SASLogon
- /credentials
- /identities

Manage Output Data Objects for REST Web Service Tasks

To associate a data object with a result field in the response, complete these steps:

- 1 Click . The Add an Output Data Object window appears.
- 2 From the **Output type** drop-down list, select **Data object**.
- 3 From the **Data object** drop-down list, select the name of the data object, or select **Create a new data object**. For more information, see [“Create a Data Object” on page 20](#).
- 4 Enter the path expression that is required to extract the value of the output parameter from the response.

Enter the name of the result field in the HTTP response that contains the information that you want to store in the data object. If the output type is JSON, specify the name using JSON path expression syntax. If the output type is XML, specify the name using a XPath expression.

- 5 Specify whether workflow processing continues if a null response is received. If you select **True** from the **Optional** drop-down list and a null response is received, then the workflow continues, using the default value or any other processing that you defined. If you select **False** from the **Optional** drop-down list and a null response is received, then the workflow fails.
- 6 Click **OK**.

To associate a transient variable with a result field in the response, complete these steps:

- 1 Click . The Add an Output Data Object window appears.
- 2 From the **Output type** drop-down list, select **Transient variable**.
- 3 Specify the name of the transient variable that will store the output.
- 4 Enter the path expression that is required to extract the value of the output parameter from the response.

Enter the name of the result field in the HTTP response that contains the information that you want to store in the transient variable. If the output type is JSON, specify the name using JSON path expression syntax. If the output type is XML, specify the name using a XPath expression.

- 5 Specify whether workflow processing continues if a null response is received. If you select **True** from the **Optional** drop-down list and a null response is received, then the workflow continues, using the default value or any other

processing that you defined. If you select **False** from the **Optional** drop-down list and a null response is received, then the workflow fails.

6 Click **OK**.

To remove or edit a data object, select the data object, and click either  or .

Invoke REST Web Service Action Example

This example illustrates a basic usage scenario for an Invoke REST web service task.

Table 4.2 Invoke REST Web Service Parameter Summary

Property Name	Description
URL	<code>/workflow-sample/rest/creditCheck?id=\${id}</code>
Method	GET
HTTP request readers	Enter <i>Content-Type</i> for the Key and <i>application/json</i> for the Value . Enter <i>Accept</i> for the Key and <i>application/json</i> for the Value .
Expected result code	<code>200</code>
Input type	Empty
HTTP response headers	<code>outputHeader</code>
Output data objects	Select rating for the Data object and enter <code>\$.rating</code> for the path expression. Select name for the Data object and enter <code>\$.name</code> for the path expression.

The following table provides the expected values of the workflow instance data objects after the credit check REST service execution.

Table 4.3 Invoke REST Web Service Results Summary

Data Object	Value
rating	575
name	John Doe
outputHeader	"123"

The Send Email Action

This action enables you to send email using the SAS Viya platform Mail Service. For more information, see [“Mail Service” in SAS Viya Platform: Configuration Properties](#).

Table 4.4 Send an Email Action Properties

Field	Description
To	Specifies the recipients for the mail message. You can list each user or group individually in the service task properties. Select Listed in a field and click  to select the users or groups. TIP To send an email to a group, you should create a data object for the group's email address. For more information, see “Create a Data Object” on page 20. You can specify a data object that contains the list of recipient IDs. Select Listed in a data object and select the data object from the drop-down list. Note: If you want to refer to a data object's value within an email notification, you must use the <code>\${data_obj_name}</code> JUEL syntax.
Subject	Specifies the subject of the email message.
Message	Specifies the message text.

Note: The sender's email address is populated by the email service with the system account information. You cannot modify the sender's email address in SAS Workflow Manager.

The Send Email Action Example

This example provides a basic usage scenario for a Send email task.

Table 4.5 *Send Email Parameter Summary*

Parameter	Value
To	list of user IDs, groups, or a data object
Subject	Email Test
Message	This is a test of the mail service.

The following figure shows an example send email action that uses a data object in the **To** field. For more information, see [“Create a Data Object” on page 20](#).

Action

Action type:

Send email

**FIELDS**

To: *

Listed in a data object



Data object:

Username



Subject: *

Email for \$(Username)

Message: *

Test message for \$(Username)

The Send Notification Action

This action enables you to send separate notifications to individual users or groups using the SAS Viya platform Notification Service. For more information, see [“Notifications Service” in SAS Viya Platform: Configuration Properties](#).

Note: The notification channel is determined by your preference settings. Supported channels include store, email, and, SMS text messages. The portal retrieves stored messages.

Table 4.6 Send Notifications Properties

Field	Description
To	Specifies the recipients for the message. You can list each recipient or group individually in the service task properties, specify a data object that contains the list of recipients, or notify only task owners. When you select Recipients or group recipients, you add users or groups by clicking . When you select Listed in an object, select Users or Group and select the data object from the drop-down list. Note: If the task is unclaimed, the message goes to all potential owners. If the task is claimed, the message goes only to the owner who claimed the task.
Template	Specifies the notification template that you want to use for the notification. For more information about creating a notification template, see “Create a Notification Template” on page 56 .
Template mapping variables	Specifies the template variable to data object mapping.

The Send Notification Action Example

This example provides a basic usage scenario for a Send notification task.

Table 4.7 Send Workflow Notification Parameter Summary

Parameter	Value
To	<i>user IDs</i>
Template	<i>SAS_Email_HTML_Message</i>
Template mapping variables	Select DESC for the Template Variable and enter Testing 123 for the Data Object field text. Select TO for the Template Variable and enter <i>dear sir</i> for the Data Object field text. Select MSG for the Template Variable . For the Data Object field, enter <i>This is a test of the notification service</i>.

The Send Group Notification Action

This action enables you to send a notification to individual users and groups where all recipients are addressed together in a single message. You can also copy additional users and groups.

Table 4.8 *Send Group Notifications Properties*

Field	Description
To	Specifies the recipients for the mail message. You can list each recipient or group individually in the service task properties, specify a data object that contains the list of recipients, or notify only task owners. When you select Recipients or group recipients, you add users or groups by clicking . You can also add users or group to the CC and BCC fields. When you select Listed in this object, select User or Group and select the data object from the drop-down list. You can also add data objects to the CC users, CC group, BCC users, and BCC group fields. Note: If the task is unclaimed, the message goes to all potential owners. If the task is claimed, the message goes only to the owner who claimed the task.
Template	Specifies the notification template that you want to use for the notification. For more information about creating a notification template, see “Create a Notification Template” on page 56.
Template mapping variables	Specifies the template variable to data object mapping.

Note: Email is the only delivery channel supported by using the group notification policy.

The Send Group Notification Action Example

This example provides a basic usage scenario for a Send group notification task.

Table 4.9 Send Workflow Group Notification Parameter Summary

Parameter	Value
To	Recipients or group recipients
Users and groups	user IDs
CC	Application Administrators
BCC	SAS Administrators
Template	SAS_Email_HTML_Message
Template mapping variables	Select DESC for the Template Variable and enter <i>Testing 123</i> for the Data Object field text. Select TO for the Template Variable and enter <i>dear sir or madam</i> for the Data Object field text. Select MSG for the Template Variable . For the Data Object field, enter <i>This is a test of the group notification service.</i>

The Invoke Job Execution Service Action

This action enables you to execute code that has been registered with the SAS Viya platform Job Execution service. A single job can execute a single task or multiple tasks bundled together.

Table 4.10 Invoke Job Execution Service Properties

Field	Description
Job definition	Specifies the available job definitions. Select the Job definition from the drop-down list.
Argument	Specifies the arguments defined for the selected job definition. You must specify a value for each job definition if no default value is provided. When the job definition has been selected, you can view the defined arguments. Select the argument and click  to specify the arguments value.
Results	Specifies the output values and associated data objects. Click  to open the Add a Result Data Object window. Select Data object and select a data object from drop-down list. You can also select and enter a transient variable. You must specify a Result name. Click OK to close the window. To remove or edit a result, select the result and click either  or .
Result state	Specifies the data object used to store the Job Execution service status. You must specify a Result state. Click  to open the Result State window. Select Data object and select the data object from the drop-down list. Note: Only data objects with a character type are listed in the Data object drop-down list. You can also select and enter a transient variable. Click OK to close the window.

Note: In order for the job definition to be displayed in the **Job definition** list, you must define the job definition with the appropriate access for all registered SAS applications or with SAS Workflow Manager.

Job definitions created using the SAS Job Execution Web Application should have the required authorization. For more information, see [“Overview” in SAS Job Execution Web Application: User’s Guide](#).

IMPORTANT The job definition must contain a parameter named `_contextName`. You must assign a valid compute context to this parameter at design time.

A user account must be configured for a workflow client in order to make service task calls within a definition. This account must be an LDAP account and also a host user account for invoking job executions. This account should be configured by the solution at start-up, or by the system administrator before the creation of any instances. For more information, see [“Configure the Workflow Client” in SAS Workflow Manager: Administrator’s Guide](#).

The Invoke Job Execution Service Action Example

This example provides a basic usage scenario for an Invoke Job Execution service task.

Table 4.11 *Submit JES Job Parameter Summary*

Parameter	Value
Job definition	<i>JobTest</i>
Argument	Select the <code>var</code> data object for the Name, and select the <code>\$category [1]</code> data object for the Value. Select the <code>_contextName</code> data object for the Name, and select the <code>SAS Job Execution compute context</code> data object for the Value.
Results	Select the <code>total</code> data object for the Name , and select the <code>theresult [2]</code> data object for the Value .
Result state	Select <code>jobStatus [3]</code> for the Data object .

Create and Use a Service Task Template

The template action enables you to specify a full or partial Invoke REST web service task definition and save it as a custom task template. After you create a task template, you can use it in other workflow definitions.

Create a Template

- 1 Follow the steps in [“Add a Service Task” on page 43](#) to create an Invoke REST web service task.
- 2 Click  in the object toolbar. Click **Save As**.
The Save as Template window appears.
- 3 Specify a name for the task template.
- 4 (Optional) Enter a description for the task template.
- 5 Select the associated application in the drop-down menu.

Note: The associated application that you select should be the same as the Client Identifier for the workflow definition.

- 6 (Optional) Specify additional data objects that you want to save in the template. The data objects that are used in the service task are already included. For more information, see [“Create and Manage Data Objects” on page 19](#).
- 7 Click **OK** to save the template.

TIP You can overwrite the saved template. Make the appropriate changes, click **Save As**, and save it with the same name.

Use a Template In a Workflow Definition

- 1 Drag the **Templated service task** element from the list of elements onto the workflow diagram. The Choose Template window appears.
- 2 Select a template. Perform one of the following actions:
 - a In the filter field, start entering a template name, and then select one in the filtered list. Click **Add**.
 - b Select a template from the list of all available templates and click **Add**.

All of the fields in the **Actions** pane are populated with the information that is stored in the template.

- 3 (Optional) On the **Properties** pane, edit any of the properties (for example, name and description). Changing the values for properties in this definition does not change the template or other definitions where the template is used. For more information, see [“Edit Element Properties” on page 13](#).
- 4 (Optional) Click  to save your workflow definition.

Managing Notification Templates

Create a Notification Template

Notification templates define email and message templates that are sent to users from the Send Notification and Send Group Notification actions. Notification templates can be used in any workflow definition.

- 1 On the Definitions category page, click : ⇨ **Manage notification configuration**.
The Manage Notification Configuration window appears.
- 2 Click **New notification**.
The New Notification window appears.
- 3 Specify a name for the notification template. The name can contain up to 250 characters.
- 4 Select the default locale (language).
- 5 (Optional) Enter a description for the notification template.
- 6 Click **Save**.
The Manage Notification Configuration window appears.
- 7 Compose your notification template. You can use the HTML editor, the text editor, or the SMS text message editor to write your notification.

Note: An SMS text message notification cannot contain more than 160 characters.

Note: If you select **SMS**, the notification cannot have a subject. If you are creating an SMS text message notification and you enter a value in the **Subject** text box, the **Subject** box is cleared when you click **Save**.

Your template can include variables. Enclose them within dollar signs (for example, `$name$`).

- 8 Click **Save**.

TIP Once you click **Save**, you can test your template. Click **Preview** to see what the notification looks like. Click **Send test notification** to send the notification to a valid email address.

In either case, you must provide values for any variables in your template.

Edit a Notification Template

- 1 On the Definitions category page, click : ⇒ **Manage notification configuration**.
- 2 Select a notification template. Click .
- 3 Edit the notification as needed.
- 4 Click **Save**.

Delete a Notification Template

You can delete a notification template if you created it, or if you are an administrator (that is, you are a member of the SASAdministrators group). Administrators can delete any template.

- 1 On the Definitions category page, click : ⇒ **Manage notification configuration**.
The Manage Notification Configuration window appears.
- 2 Select one or more notification templates.
- 3 Click .
- A window prompts you to confirm the deletion.
- 4 Click **Delete**.

Working with Subprocesses and Call Activities

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About Subprocesses and Call Activities

Subprocesses and call activities are ways to modularize and reuse workflow logic. Subprocesses and call activities both define workflow processes. Subprocess definitions are embedded in the main workflow definition. Call activities are defined separately and referenced from the main workflow definition. For a running workflow instance, a subprocess runs within the same process as the main process. Call activities run as separate processes.

Using subprocesses and call activities improves readability of a workflow definition and promotes consistent reuse of business logic within an organization.

Add a Subprocess

- 1 Drag the **Subprocess** element from the list of objects onto the workflow diagram.
- 2 (Optional) On the **Properties** pane, edit the name and description. For more information, see [“Edit Element Properties” on page 13](#).

- 3 Open the subprocess in a new tab. Do one of the following:
 - Click  next to the subprocess name within the workflow definition view.
 - Double-click the subprocess within the workflow definition view.
- 4 To add an element, drag the element from the list of elements onto the subprocess diagram. The **Start** and **End** nodes are automatically added to the element.

Note: A subprocess definition must contain a **Start** element and at least one **End** or **Terminate End** element.

Note: If an element in your subprocess uses a data object, you must define the data object before you can configure that element.

- 5 Define the data objects that are required by the subprocess. You define data objects in the Definition Properties window.

Note: The tab that is selected when you define a data object determines the scope of the data object. For more information, see [“About Data Objects and Scope” on page 19](#).

- 6 (Optional) Configure the **Start** element.

You configure subprocess elements by editing the element properties. For more information, see [“Edit Element Properties” on page 13](#).

- 7 Add and configure the elements required in the subprocess. For additional information, see the following topics:

- [Chapter 3, “Working with User Tasks,” on page 29](#)
- [Chapter 4, “Working with Service Tasks,” on page 41](#)
- [Chapter 5, “Working with Subprocesses and Call Activities,” on page 59](#)
- [Chapter 6, “Using Timers,” on page 65](#)
- [Chapter 7, “Using Signals,” on page 69](#)
- [Chapter 8, “Using Messages,” on page 73](#)
- [Chapter 9, “Using Gateways,” on page 77](#)
- [“Add Annotations and Associations” on page 16](#)

- 8 Add an **End** or **Terminate end** to the subprocess definition.

TIP The **End** element ends the subprocess defined by the workflow path to which it is added, but it allows its subprocesses to complete. The **Terminate end** element terminates the entire subprocess and all of its subprocesses.

- 9 Add sequence flows to connect the elements in the subprocess. To connect two elements together, click  to enable link mode, and then drag the cursor from the first element to the next.

TIP To move elements in the diagram, click  to enable move mode. You can switch between link mode () and move mode () as needed.

For more information, see [“Add Sequence Flows” on page 14](#).

- 10 (Optional) Specify conditions for the sequence flows. The conditions on sequence flows are evaluated when a workflow instance executes, and the results determine which paths are taken. For more information, see [“Add Sequence Flow Conditions” on page 15](#).
- 11 Click  to save the workflow definition.

Add a Call Activity

- 1 Drag the **Call activity** element from the list of objects onto the workflow diagram.
- 2 (Optional) On the **Properties** pane, edit the name and description. For more information, see [“Edit Element Properties” on page 13](#).
- 3 Click  to display the **Process** pane.
- 4 Select the process (workflow) definition that you want to call.
- 5 (Optional) Specify the input parameters for the call activity. Input parameters pass data object values into the call activity. Select the data objects in the current workflow (source) whose values need to be passed to the call activity process (target).
 - a Click  above the **Input parameters** table. The New Input Parameter window appears.
 - b In the **Source** field, select the data object whose value you want to pass to the call activity.
 - c In the **Target** field, select the data object in which you want to store the value of the source data object.
 - d Click **OK** to close the New Input Parameter window.

TIP To edit or remove an input parameter, select the parameter in the table and click  or .

- 6 (Optional) Specify the output parameters for the call activity. Output parameters pass data object values from the call activity.
 - a Click  above the Output Parameters table. The New Output Parameter window appears.
 - b In the **Source** field, select the data object whose value you want to pass to the current workflow.
 - c In the **Target** field, select the data object in which you want to store the value of the source data object.
 - d Click **OK** to close the New Output Parameter window.

TIP To edit or remove an output parameter, select the parameter and click  or .

Create and Use a Subprocess Template

The template action enables you to specify a full or partial subprocess task definition and save it as a custom subprocess template. After you create a subprocess template, you can use it in other workflow definitions.

Create a Subprocess Template

- 1 Follow the steps in [“Add a Subprocess” on page 59](#) to create a subprocess.
- 2 Click  in the object toolbar. Click **Save As**.
The Save As Template window appears.
- 3 Specify a name for the subprocess template.
- 4 (Optional) Enter a description for the subprocess template.
- 5 Select the associated application in the drop-down menu.

Note: The associated application that you select should be the same as the Client Identifier for the workflow definition.

- 6 (Optional) Specify additional data objects that you want to save in the template. The data objects that are used in the subprocess are already included. For more information, see [“Create and Manage Data Objects” on page 19](#).
- 7 Click **OK** to save the template.

TIP You can overwrite the saved template. Make the appropriate changes, click **Save As**, and save it with the same name.

Use a Subprocess Template In a Workflow Definition

- 1 Drag the **Templated subprocess** element from the list of elements onto the workflow diagram. The Choose Template window appears.
- 2 Select a template. Perform one of the following actions:
 - In the filter field, start entering a subprocess template name, and then select one in the filtered list. Click **Add**.
 - Select a template in the list of all available templates and click **Add**.

All of the fields in the **Actions** pane are populated with the information that is stored in the template.

- 3 (Optional) On the **Properties** pane, edit any of the properties (for example, name and description). Changing the values for properties in this definition does not change the template or other definitions where the template is used. For more information, see [“Edit Element Properties” on page 13](#).
- 4 (Optional) Click  to save your workflow definition.

Using Timers

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About Timers

You can use timers to trigger workflow activities at specific times or after specified intervals. For example, you can use timers to delay the start of a task or subprocess, cancel an activity after a specified interval, or send an email reminder at a specific time.

SAS Workflow Manager provides two different types of timers:

boundary timers

are associated with a specific activity. The timer starts when process execution arrives at the associated activity. The timer can be set to fire immediately, after a relative delay, or at a specific time. You can set boundary timers to repeat at specific intervals, or you can use them to cancel the associated activity. When the timer fires, all of the outgoing paths from the timer are followed. If the timer is set to repeat, all outgoing paths are followed every time the timer fires. If the associated activity completes before the timer fires, the timer is canceled.

Note: A boundary timer must have at least one outgoing sequence flow.

intermediate timers

are independent elements of a workflow. They are not attached to a specific activity. The timer starts when process execution arrives at the timer. The timer can be set to fire immediately, after a relative delay, or at a specific time. Intermediate timers cannot be set to repeat, and they cannot be used to cancel an activity. When the timer fires, all of the outgoing paths from the timer are followed.

If you set a timer to fire at a specific time and that time has already passed when the timer starts, then the timer fires immediately.

Note: Datetime type data objects without default values are initialized to the current datetime at process start because data objects cannot be null. Ensure that the default values for Datetime type data objects are set properly before the timer is accessed.

Add a Boundary Timer

- 1 Drag the **Boundary timer** from the list of events onto the task with which you want to associate the timer.

TIP You can also highlight the element and click  in the properties pane. Click  above the boundary timer table to configure a timer.

TIP You can also click  on the element or right-click the element to add a new boundary timer by using the current elements pop-up menu.

- 2 (Optional) On the **Properties** pane, edit the name and description. For more information, see [“Edit Element Properties” on page 13](#).
- 3 Click  to display the **Timer** property pane. The **Timer firing time** value indicates the next firing time as determined by the timer settings. If the timer execution is set to **When the timer is reached**, then the **Timer firing time** value is displayed as `(recurrence)/PT(recurrence interval or delay)`. For example, a timer with a delay of 2 seconds shows the **Timer firing time** value as `PT2S` and a timer with a maximum recurrence of 3 and a recurrence interval of 10 seconds shows the value as `R3/PT10S`. If the timer execution is set to **Specified by a data object**, the **Timer firing time** value is displayed as the data object that you selected.
- 4 (Optional) Clear the **Cancel the activity** check box if you do not want the timer to cancel the activity when it fires.
- 5 Set the time at which you want the timer to fire.
 - To fire the timer immediately when the workflow reaches the activity, select **When the timer is reached**.
 - To fire the timer after a delay, select the delay under **Delay**. This delay is relative to the time at which the timer started.

- To exclude specific dates for the boundary timer, add a business calendar. Select the business calendar from the drop-down list under **Excluded dates**. Click **View item** to view the selected calendar.

Note: Only a SAS Workflow Administrator can create, edit, or delete business calendars. For more information, see [“Managing Business Calendars” in SAS Workflow Manager: Administrator’s Guide](#).

- To fire at a specific time, select **Specified by a data object**, and select the data object or select **Create a new data object** from the drop-down list. The data object is evaluated when the associated task starts.

Note: If you set a timer to fire at a specific time and that time has already passed when the timer starts, then the timer fires immediately.

6 (Optional) Set the timer to repeat.

- a Clear the **This event occurs only once** check box.

Note: You must clear the **Cancel the activity** check box in order to perform this action.

- b Specify the recurrence interval.

- c (Optional) To exclude specific dates for the recurring boundary timer, add a business calendar. Select the business calendar from the drop-down list under **Excluded dates**. Click **View item** to view the selected calendar.

Note: Only a SAS Workflow Administrator can create, edit, or delete business calendars. For more information, see [“Managing Business Calendars” in SAS Workflow Manager: Administrator’s Guide](#).

- d Specify when you want the timer to stop repeating.

- To repeat the timer as long as the associated activity is active, select **No end date or time**.
- To specify a maximum number of occurrences, select a value for **Maximum number of occurrences, including the first**.
- To specify a specific end date and time, select **Repeat until the date and time specified by this data object**, and select the data object.

Note: If you specify both a maximum number of occurrences and an end date and time, the timer stops firing when the first of those conditions is met.

Add an Intermediate Timer

- 1 Drag the **Intermediate timer** from the list of elements onto the workflow diagram.

TIP You can also click  on the element or right-click the element to add a new intermediate timer using the current elements pop-up menu.

- 1 (Optional) On the **Properties** pane, edit the name and description. For more information, see [“Edit Element Properties” on page 13](#).
- 2 Click  to display the **Timer** property pane. In the **Timer** property pane, the **Timer firing time** value indicates the next firing time as determined by the timer settings. If the timer execution is set to **When the timer is reached**, the **Timer firing time** value is displayed as **PT(delay)**. For example, a timer with a delay of 2 seconds shows the **Timer firing time** value as **PT2S**. If the timer execution is set to **Specified by a data object**, the **Timer firing time** value is displayed as the data object you selected.
- 3 Set the time at which you want the timer to fire.
 - To fire the timer immediately when the workflow reaches the activity, select **When the timer is reached**.
 - ☐ To fire the timer after a delay, select the delay under **Delay**. This delay is relative to the time at which the timer started.
 - ☐ To exclude specific dates for the boundary timer, add a business calendar. Select the business calendar from the drop-down list under **Excluded Dates**. Click **View item** to view the selected calendar.

Note: Only a SAS Workflow Administrator can create, edit, or delete business calendars. For more information, see [“Managing Business Calendars” in SAS Workflow Manager: Administrator’s Guide](#).

- To fire the timer at a specific time, select **Specified by a data object**, and select the data object or select **Create a new data object** from the drop-down list. The data object is evaluated when the instance that contains the timer starts.

Note: If you set a timer to fire at a specific time and that time has already passed when the timer starts, then the timer fires immediately.

Using Signals

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About Signals

Signals indicate to a workflow that an external event has occurred and can affect the execution of the workflow. Signals are events of global scope. When a signal is received, the recipients proceed with no reply necessary. In SAS Workflow Manager, signal events provide greater flexibility, automation, and simplicity in the workflow.

SAS Workflow Manager provides two different types of signal events:

boundary signals

catch signals if they are received during processing of the activity which the boundary signal is placed on. If a boundary signal is triggered, it proceeds with the outgoing sequence flow. They do not halt processing of the activity. Boundary signals can optionally cancel the activity if triggered.

intermediate signals

are independent events in a workflow. They are not attached to a specific activity. When the intermediate signal event is entered, the workflow is halted and does not proceed until the intermediate signal is triggered by a broadcasted signal matching the signal reference.

Add a Boundary Signal

- 1 Drag the **Boundary signal** from the list of events onto the task with which you want to associate the timer.

TIP You can also highlight the activity and click  in the properties pane. Click  in the **Signal** property pane to configure a new boundary signal.

TIP You can also click  on the task or right-click the task to add a new boundary signal by using the current elements pop-up menu.

- 2 (Optional) On the **Properties** pane, edit the name and description. For more information, see [“Edit Element Properties” on page 13](#).
- 3 Click  to display the **Signal** property pane.
- 4 (Optional) Clear the **Cancel the activity** check box if you do not want the signal to cancel the workflow activity that the boundary signal is placed on when it fires.
- 5 Specify the signal reference. This specifies the name of the signal that the boundary signal event is waiting for.

Note: Boundary signals must have an outgoing sequence flow. When the signal is received by a boundary signal, it executes that outgoing sequence flow.

- 6 Click  to save the workflow.

TIP To edit the properties of a boundary signal later, right-click the boundary signal and select **Edit**. The property pane automatically displays the properties for the boundary signal, and you can edit the signal reference.

To delete a boundary signal later, right-click the boundary signal and select **Delete**.

Add an Intermediate Signal

- 1 Drag the **Intermediate signal** from the list of events onto the workflow diagram.

TIP You can also click  on a task or right-click a task to add a new intermediate signal by using the current elements pop-up menu.

- 2 (Optional) On the **Properties** pane, edit the name and description. For more information, see [“Edit Element Properties” on page 13](#).
- 3 Click  to display the **Signal** property pane.
- 4 Specify the signal reference. When the workflow arrives at the intermediate signal, it halts until the signal with this signal reference is broadcast.
- 5 (Optional) Specify prompts for the intermediate signal. Click  to display the **Prompts** property pane. For more information, see [“Managing Prompts” on page 36](#).
- 6 Click  to save the workflow.

TIP To edit the properties of an intermediate signal later, click the intermediate signal. The property pane automatically displays the properties for the intermediate signal, and you can edit the signal reference and prompts.

To delete an intermediate signal later, right-click the intermediate signal and select **Delete**.

Using Messages

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About Messages

Messages provide a way for applications to communicate with the workflow. Messages are character strings that are sent to a running workflow instance by another process. The process sends the message through the workflow service.

You can use messages to trigger workflow activities and to cancel activities.

SAS Workflow Manager provides two different types of messages:

boundary messages

are associated with a specific activity. You can use boundary messages to cancel the associated activity. When process execution arrives at the associated activity, the workflow waits for the message to arrive before either proceeding with the process or canceling the activity. When the message arrives, all of the outgoing paths from the message event are followed. If the associated activity completes before the message arrives, the message event is canceled.

intermediate messages

are independent elements of a workflow. They are not attached to a specific activity. When process execution arrives at the intermediate message event, the workflow waits for the message to arrive. When the message arrives, all of the outgoing paths from the message event are taken.

Note: A message reference is required for boundary messages. You can create a boundary message without a message reference value, but the workflow definition fails when you try to create a new version. A workflow definition is not fully validated until it is saved as a version.

Add a Boundary Message

- 1 Drag the **Boundary message** from the list of events onto the task with which you want to associate the message.

TIP You can also highlight the activity and click  in the properties pane. Click  in the **Message** property pane to configure a new boundary message.

- 2 (Optional) On the **Properties** pane, edit the name and description. For more information, see [“Edit Element Properties” on page 13](#).
- 3 Click  to display the **Message** property pane.
- 4 (Optional) Clear the **Cancel the activity** check box if you do not want to cancel the activity when the message arrives.
- 5 Enter the **Message reference**. The message reference is the label used to identify the appropriate character string that is sent to the running workflow.

TIP When you add a boundary message, the property pane automatically displays the properties for the boundary message. To delete or edit the properties of a boundary message later, right-click the boundary message, and select **Edit** or **Delete**.

You can also highlight the activity and click  to access the properties pane. Select the boundary message, and click  to edit or  to remove.

Add an Intermediate Message

- 1 Drag the **Intermediate message** from the list of elements onto the workflow diagram.
- 2 (Optional) On the **Properties** pane, edit the name and description. For more information, see [“Edit Element Properties” on page 13](#).
- 3 Click  to display the message properties pane.
- 4 Enter the **Message reference**. The message reference is the label used to identify the appropriate character string that is sent to the running workflow.

TIP To edit the properties of an intermediate message later, click the intermediate message. The property pane automatically displays the properties for the intermediate message, and you can edit the message reference.

To delete an intermediate message later, right-click the intermediate message and select **Delete**.

Using Gateways

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About Gateways

Gateways enable you to control or synchronize how sequence flows converge and diverge in a workflow. For example, your workflow might have five workflow paths that converge to a gateway that has two outgoing paths, and the outgoing paths must all be taken at the same time. You can use a parallel gateway to control this flow.

If you do not need to control how paths diverge and converge, then you do not need to use a gateway. For more information about controlling the flow of a workflow, see [Chapter 11, “Understanding Workflow Patterns,” on page 87](#).

[Table 9.1 on page 77](#) describes the gateway types provided by SAS Workflow Manager.

Table 9.1 Gateway Types

Type	Action when used to fork paths	Action when used to merge paths
Parallel	The workflow takes all of the outgoing paths. You cannot specify conditions for outgoing paths on a parallel gateway. For more information, see “Using Parallel Gateways (AND-split)” on page 90 .	The workflow waits for all of the incoming paths to arrive before it allows the workflow process to continue executing. If any incoming path is not active, the workflow becomes deadlocked. For more information, see “Merging with Parallel Gateways (Join)” on page 95 .

Type	Action when used to fork paths	Action when used to merge paths
Exclusive	The workflow takes only one outgoing path. The path that is taken is the first path whose condition evaluates to True. For more information, see “Using Exclusive Gateways (XOR)” on page 90 .	You should not use exclusive gateways to merge paths. When an exclusive gateway is used to merge paths, it behaves like a simple merge. For more information, see “Merging Alternate Paths (Simple Merge)” on page 94 .
Inclusive	The workflow either takes all of the outgoing sequence flows whose conditions evaluate to True, or it takes the default path if none evaluate to True and you have designated a default path. For more information, see “Using Inclusive Gateways (OR-split)” on page 91 .	The workflow waits for all of the active incoming paths to arrive before it allows the workflow process to continue executing. If at least one of the incoming paths is active, the gateway allows the workflow process to continue. For more information, see “Merging with Inclusive Gateways (Inclusive Merge)” on page 96 .
Event-based	The workflow takes only one outgoing path. The path that is taken is the first path that is triggered by an event. For more information, see “Using Event-Based Gateways” on page 93 .	You should not use event-based gateways to merge paths.

Add a Parallel Gateway

- 1 Drag the **Parallel gateway** object from the list of elements onto the workflow diagram.
- 2 (Optional) On the **Properties** pane, edit the name and description. For more information, see [“Edit Element Properties” on page 13](#).
- 3 Click  to enable link mode, and connect the appropriate workflow elements to the gateway.

Add an Exclusive Gateway

- 1 Drag the **Exclusive gateway** object from the list of elements onto the workflow diagram.

- 2 (Optional) On the **Properties** pane, edit the name and description. For more information, see [“Edit Element Properties” on page 13](#).
- 3 Click  to enable link mode, and connect the appropriate workflow elements to the gateway.
- 4 Enter conditional expressions for the outgoing sequence flows. For each sequence flow that requires a conditional expression, complete the steps listed in [“Add Sequence Flow Conditions” on page 15](#).
- 5 Select the gateway.
- 6 Click  to display the **Conditions** property pane.
- 7 Verify that the conditional expressions are listed in the order in which you want them to be evaluated when the workflow is running. For more information, see [“Using Exclusive Gateways \(XOR\)” on page 90](#).

Add an Inclusive Gateway

- 1 Drag the **Inclusive gateway** object from the list of objects onto the workflow diagram.
- 2 (Optional) On the **Properties** pane, edit the name and description. For more information, see [“Edit Element Properties” on page 13](#).
- 3 Click  to enable link mode, and connect the appropriate workflow elements to the gateway.
- 4 Enter conditional expressions for the outgoing sequence flows. For each sequence flow that requires a conditional expression, complete the steps listed in [“Add Sequence Flow Conditions” on page 15](#).
- 5 Select the gateway.
- 6 Click  to display the **Conditions** property pane.
- 7 Verify that the conditional expressions are listed in the order in which you want them to be evaluated when the workflow is running. For more information, see [“Using Inclusive Gateways \(OR-split\)” on page 91](#).
- 8 (Optional) To designate a default sequence flow, select the row for the sequence flow, and click . SAS Workflow Manager displays a check mark (✓) next to the default sequence flow. For more information, see [“Using Inclusive Gateways \(OR-split\)” on page 91](#).

TIP To remove the default designation, select the default and click .

Add an Event-Based Gateway

- 1 Drag the **Event-based gateway** object from the list of elements onto the workflow diagram.
- 2 (Optional) On the **Properties** pane, edit the name and description. For more information, see [“Edit Element Properties” on page 13](#).
- 3 Drag either an **Intermediate message**, **Intermediate signal**, or **Intermediate timer** element onto the workflow diagram.
- 4 Click  to enable link mode, and connect the appropriate workflow elements to the gateway.
- 5 (Optional) Select the gateway and then click  to display the **Endpoints** property pane.

Localizing Workflow Text

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About Languages and Workflows

You can specify localized text for the following items in a workflow definition:

- names and descriptions of definitions, subprocesses, user tasks, and service tasks
- names of data objects
- display text for prompts

You can also specify a default language for use when you are working with a workflow definition in SAS Workflow Manager.

Localized Text in SAS Workflow Manager

When you are defining a workflow in SAS Workflow Manager, the text that you see is determined as follows:

- 1 If localized text is defined for your current locale, then you see text in the language for your locale.
- 2 For any element that does not contain localized text for your current locale, you see the localized text for the default language that is specified in the definition.
- 3 If an element does not contain localized text for the default language specified in the workflow definition, then you see English text for the **en-US** locale.

Note: Date and number formats are localized for the locale that was in use when the workflow definition was created. If you import the definition into an environment that is using a different locale, then the localized date and number formats remain localized for the original locale.

Localized Text in a Workflow Instance

For users that are working with a running instance of a workflow, the text that the users see is determined by the locale that is specified in the `ACCEPT-LANGUAGE` header field. This header field is in the request that is sent to the workflow service by the client application.

- 1 If localized text is defined for the locale that is specified in the `ACCEPT-LANGUAGE` header, then the user sees the text for that locale.
- 2 For any element that does not contain localized text for that locale, the user sees the text that was entered into SAS Workflow Manager when the workflow was defined.

Set the Default Language

TIP The default language specified in the workflow definition is used only in SAS Workflow Manager. The default language does not affect what the user sees in a running workflow instance.

- 1 Click the tab for the main process.
- 2 Click  in the object toolbar, and select **Languages**. The Localize Text window appears.
- 3 Select the default language, and click **OK**.

Localize Definition Names and Descriptions

- 1 Click the tab for the main process.
- 2 Click , and select **Languages**. The Localize Text window appears.
- 3 Complete these steps for each language:
 - a Click  above the **Name** table. The Localized Text window appears.
 - b Select the language.
 - c Enter the definition name in the selected language in the text field.
 - d Click **OK**.
- 4 Repeat [Step 3](#) for the **Description**.

TIP Only localized definition names and descriptions that are not in the set default language can be edited or removed from the localized text window. Select the definition name or description, click  to edit or  to remove.

Localize Subprocess, User Task, and Service Task Information

- 1 Select the element for which you want to localize the name and description.
- 2 Click  to display the **Localization** property pane.
- 3 Complete these steps for each language:
 - a Click  above the **Name** table. The Localized Text window appears.
 - b Select the language.
 - c Enter the element name in the selected language in the text field.
 - d Click **OK**.
- 4 Repeat [Step 3](#) for the **Description**.

TIP Only localized definition names and descriptions that are not in the set default language can be edited or removed from the localized text window. Select the definition name or description, click  to edit or  to remove.

Localize Data Object Names

- 1 Click the tab for the main process or a subprocess, depending on whether you want to localize global or local data object names. For more information, see [“About Data Objects and Scope” on page 19](#).
- 2 Click . The Definition Properties window appears.
- 3 For each data object, complete these steps for each language:
 - a Select the data object and click **Localize**. The Localize Text window appears.
 - b Click . The Localized Text window appears.
 - c Select the language.
 - d Enter the data object name in the selected language in the text field.
 - e Click **OK** to close the Localized Text window.

- 4 Click **OK** to close the Definition Properties window.

TIP Only localized data objects that are not in the set default language can be edited or removed from the localized text window. Select the data object, click  to edit or  to remove.

Localize Prompts

- 1 Select the user task or Start element for which you want to localize the prompts.
- 2 Click  to display the **Prompts** property pane.
- 3 For each prompt, complete these steps for each language:
 - a Select the prompt and click **Localize**. The Localize Text window appears.
 - b Click . The Localized Text window appears.
 - c Select the language.
 - d Enter the prompt text in the selected language in the text field..
 - e Click **OK** to close the Localized Text window.

TIP Only localized prompts that are not in the set default language can be edited or removed from the localized text window. Select the prompt, click  to edit or  to remove.

Understanding Workflow Patterns

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About Workflow Patterns

Only the most basic workflows can be represented as a single, sequential flow. Most workflows contain combinations of activities, events, and gateways. A workflow definition defines the possible paths through these elements and is based on your organization's business processes. The actual path that is taken by a

workflow instance is determined at run time by the outcome of each activity and by the evaluation of conditional expressions associated with sequence flows.

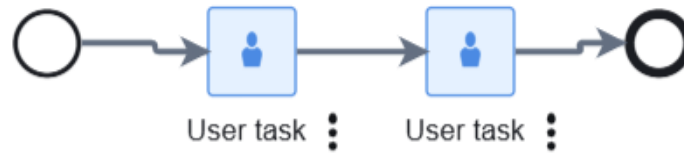
A workflow definition typically uses one or more workflow patterns. A workflow pattern is a process construct that defines how a workflow behaves. For example, many workflows have a decision point where the main workflow path diverges, and alternative paths are available through the workflow. Alternate paths might be mutually exclusive, or they might run in parallel. You can control how the alternate paths are followed by using conditional expressions. How the workflow behaves depends on the workflow patterns that are modeled in the definition.

Table 11.1 *Types of Workflow Patterns*

Type	Description
Sequential	The workflow defines a single path through the workflow elements. For more information, see “Sequential Flow” on page 88 .
Divergent	The workflow defines a path that splits into multiple alternative paths. If more than one path is executed, they are executed in parallel. You can use conditional sequence flows or gateways to implement divergent patterns. For more information, see “Divergent Patterns” on page 89 .
Convergent	The workflow defines multiple paths that merge at the same element. For more information, see “Convergent Patterns” on page 94 .
Termination	A workflow path reaches an End or Terminate End element. For more information, see “Termination Patterns” on page 97 .
Cancellation	An activity or process is canceled, and the workflow does not reach an End or Terminate End element. For more information, see “Cancellation Patterns” on page 99 .
Multiple Instance	A separate instance of a task is created for each participant. For more information, see “Multiple Instance Pattern” on page 97 .

Sequential Flow

The most basic workflows are represented as a single sequential flow. Each task is completed in sequence, and no conditions are associated with the sequence flows. However, most workflows are not this simple.

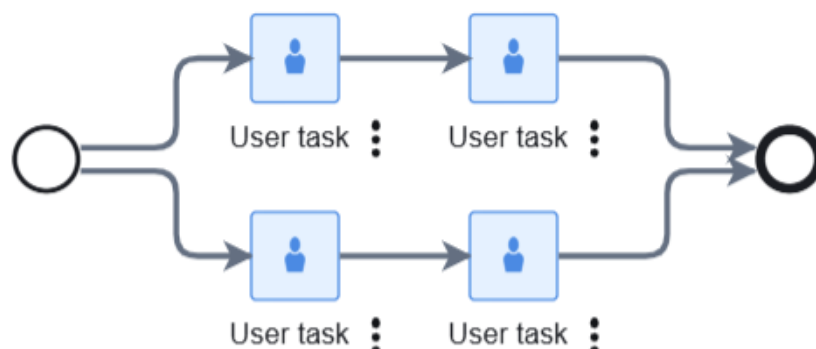
Figure 11.1 Sequential Workflow Diagram

Divergent Patterns

Using Alternate Paths

Some workflows have decision points and alternate paths, and each path represents a different business case. For example, a specific path might depend on the value of a data object or on a participant's response to a prompt. Suppose you have the workflow shown in [Figure 11.2 on page 89](#). You can define a prompt for the Start node. Then, you can define conditional expressions for the outgoing sequence flows on the Start node that evaluate the participant's answer. Depending on the results, the running workflow might take the path to Task 1, the path to Task 3, or both paths. If it takes both paths, then Tasks 1 and 3 are executed in parallel.

If neither of the conditional expressions defined for the sequence flows on the Start node evaluate to True, the workflow process terminates.

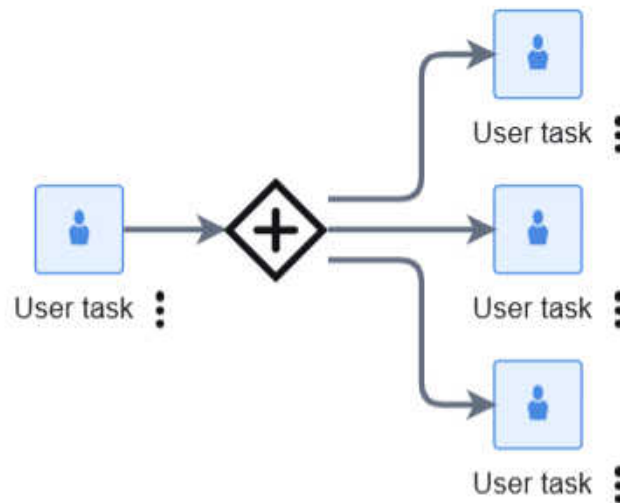
Figure 11.2 Workflow with Alternate Paths

An alternative to simple alternate paths is an inclusive gateway. For more information, see [“Using Inclusive Gateways \(OR-split\)” on page 91](#).

Using Parallel Gateways (AND-split)

You can use a parallel gateway when the workflow needs to take multiple paths at the same time. You cannot associate conditions with the outgoing sequence flows from a parallel gateway. All of the outgoing paths are followed, and the subsequent activities run in parallel. In the workflow shown in [Figure 11.3](#), Tasks 2, 3, and 4 all run in parallel

Figure 11.3 Divergent Workflow with Parallel Gateway



For more information, see [“Add a Parallel Gateway” on page 78](#).

Using Exclusive Gateways (XOR)

With an exclusive gateway, only a single outgoing path is executed. The path that is chosen is based on the conditions that you specify for the outgoing sequence flows. The conditions are evaluated in the order in which you define them. Only the sequence flow for the first condition that evaluates to True is executed.

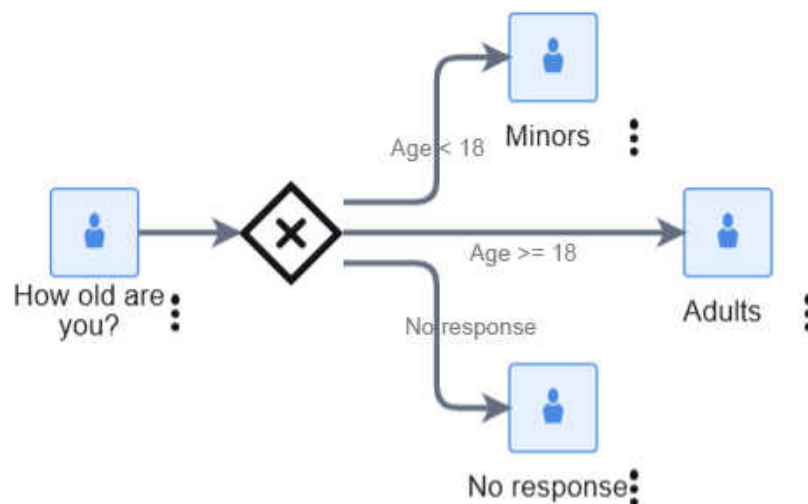
A sequence flow with no associated conditions becomes the default sequence flow. If the conditions on all of the other sequence flows evaluate to False, the default sequence flow is taken. When you specify the order in which the sequence flows are evaluated, the default sequence flow must be the last one in the list.

For example, suppose the workflow shown in [Figure 11.3](#) defines an optional prompt for the task Enter Order that prompts the participant to enter their age. The conditions that are defined for outgoing sequence flows on the gateway are as follows:

Condition	Endpoint
age<18	Minors
age>=18	Adults
	No Response

If the prompt is optional and the user chooses not to enter their age, the workflow sets the value to zero and takes the path to the No Response task.

Figure 11.4 Divergent Workflow with Exclusive Gateway



For more information, see [“Add an Exclusive Gateway” on page 78](#).

Using Inclusive Gateways (OR-split)

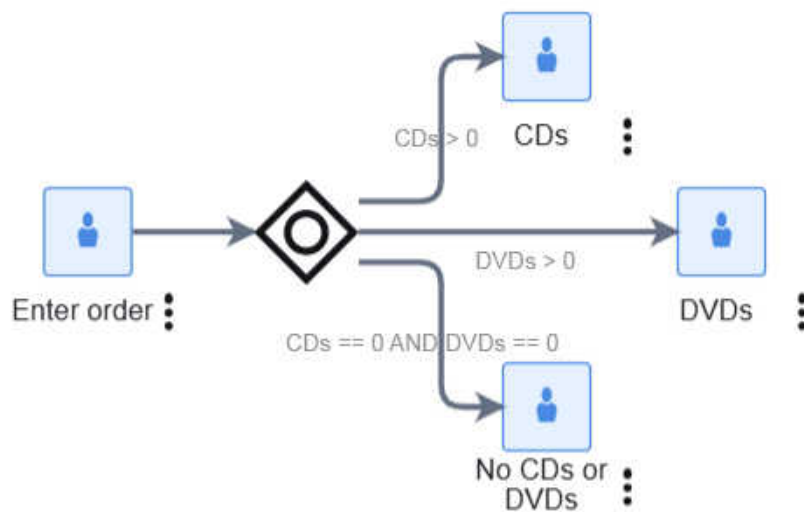
For an inclusive gateway, all of the sequence flows whose conditions evaluate to True are taken. A sequence flow with no associated conditions always evaluates to True, and that path is always taken. (A sequence flow with no associated condition is an *unconditional* sequence flow.)

For example, suppose Task 1 in [Figure 11.5 on page 92](#) requires that the participant enter a customer’s order information. The customer could order CDs and DVDs, which require special handling. The following conditions are defined for outgoing sequence flows on the gateway:

Condition	Endpoint
numCDs>0	CDs
numDVDs>0	DVDs
numDVDs==0 AND numCDs==0	No CDs or DVDs

If a customer orders CDs, the CDs task starts. If the customer orders DVDs, the DVDs task starts. If the customer orders both CDs and DVDs, then both of those tasks start. If the customer's order does not contain any CDs or DVDs, then only the default task (No CDs or DVDs) starts.

Figure 11.5 Divergent Workflow with Inclusive Gateway



When you configure the gateway, you specify the order in which the sequence flows are evaluated. You can also designate a default sequence flow. The default sequence flow is taken only if all of the other paths evaluate to False. If any of the sequence flows are unconditional, you can designate one of the unconditional sequence flows only as the default flow. If all of the sequence flows have conditions, then you can designate any of the sequence flows as the default flow.

For example, suppose you have an inclusive gateway that diverges to three tasks. Only one of the outgoing sequence flows has a condition, and the first of the unconditional sequence flows is designated as the default flow. The last sequence flow in the list is unconditional. The table for this gateway is shown in [Table 11.2 on page 93](#).

If the condition on the first sequence flow evaluates to False, this workflow takes only the path to Task 3.

Table 11.2 Inclusive Gateway Conditions That Execute Only Task 3

Condition	Endpoint	Default	Path Is Taken
Evaluates to False	Task 1	No	No
No condition (always True)	Task 2	Yes	No
No condition (always True)	Task 3	No	Yes

For more information, see [“Add an Exclusive Gateway” on page 78](#).

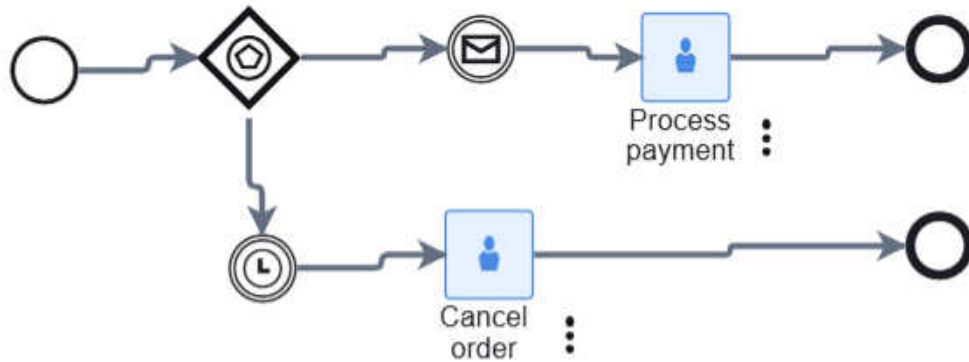
Using Event-Based Gateways

An event-based gateway determines the path that a process takes, based on events. Each outgoing sequence flow must be connected to an intermediate timer or message event. When a process reaches an event-based gateway, the execution is suspended, and for each outgoing sequence flow, an event subscription is created. The process follows the path for the event that occurs first. No other events in the gateway can be triggered after the path is determined.

For example, suppose the workflow shown in [Figure 11.6](#) illustrates an event-based gateway when the payment on an order has failed. The process can take one of two paths. In the first path, an intermediate message event occurs and the payment details are updated. Then, the process proceeds to a service task where the payment is processed. In the second path, an intermediate timer event is triggered. If the payment details are not updated within two hours, the process proceeds to a service task where the order is canceled. Whichever intermediate event occurs first determines which path the process takes. The conditions that are defined for outgoing sequence flows on the gateway are described in the following table.

Table 11.3 Conditions and Endpoints for a Divergent Workflow with an Event-Based Gateway

Intermediate catching event	Condition	Endpoint
Intermediate Message	Payment details are updated	Payment is processed
Intermediate Timer	If there is no response after two hours, proceed with cancellation	Order is canceled

Figure 11.6 Divergent Workflow with Event-Based Gateway

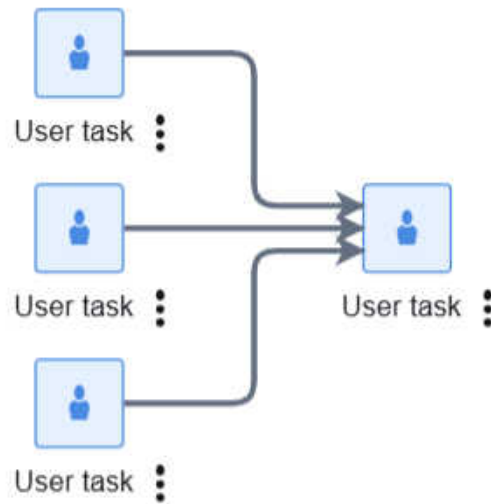
Convergent Patterns

Merging Alternate Paths (Simple Merge)

It is possible to merge workflow paths by doing a simple merge in which multiple outgoing paths merge at the same element. Simple merges do not use a gateway.

Simple merges are most appropriate when the paths are mutually exclusive.

Suppose you have the workflow shown in [Figure 11.7 on page 95](#). If the paths are mutually exclusive, the workflow takes only one of the incoming paths to Task 5, and only one instance of Task 5 is created. If the paths are not mutually exclusive and the workflow takes more than one of the incoming paths, a new instance of Task 5 is created for each path taken.

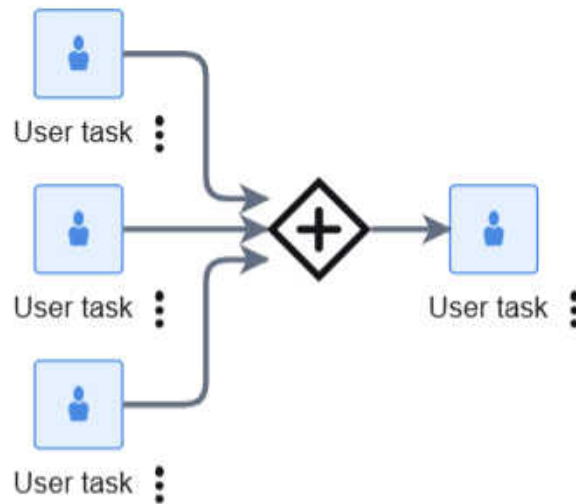
Figure 11.7 Merging Paths without a Gateway

TIP To run multiple occurrences of the same task, you can specify that a new instance of the task is created for each participant. For more information, see [“Enable Multiple Participants to Complete a Task” on page 32](#).

Note: A special situation exists if the paths converge at a Terminate End element. For more information, see [“Explicit Termination” on page 98](#).

Merging with Parallel Gateways (Join)

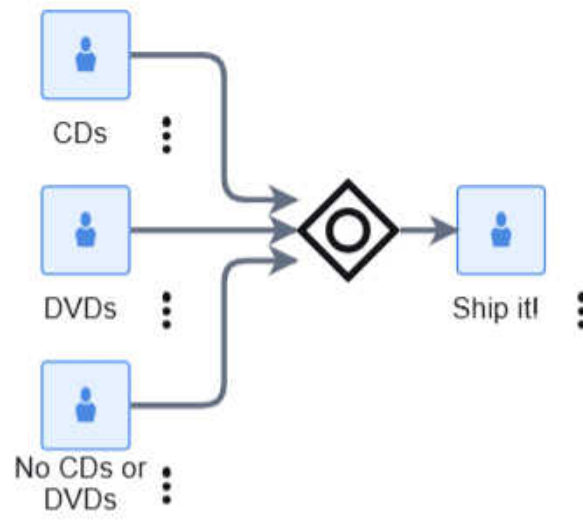
When you merge paths by using a parallel gateway, the gateway waits for all of the incoming tasks to complete before it allows the workflow process to continue executing. (The Workflow Service knows which tasks are active and when they complete.) In [Figure 11.8 on page 96](#), the gateway waits for Tasks 2, 3, and 4 to complete before allowing Task 5 to start.

Figure 11.8 Convergent Workflow with a Parallel Gateway

For more information, see [“Add a Parallel Gateway” on page 78](#).

Merging with Inclusive Gateways (Inclusive Merge)

An inclusive gateway waits for all of the active tasks to complete. (The Workflow Service knows which tasks are active and when they complete.) For example, in [Figure 11.9 on page 97](#) if only the paths from the CDs and DVDs tasks to the gateway are taken, the gateway triggers the Ship it! task when the CDs and DVDs tasks are complete. The workflow does not wait for the No CDs or DVDs task to run.

Figure 11.9 Convergent Workflow with an Inclusive Gateway

For more information, see [“Add an Inclusive Gateway” on page 79](#).

Multiple Instance Pattern

When you assign multiple participants to a user task, you can specify that a separate instance of the task is created for each participant. This pattern is referred to as Multiple Instances with a priori Design-Time Knowledge. For more information, see [“Enable Multiple Participants to Complete a Task” on page 32](#).

Note: A simple merge as described in [“Merging Alternate Paths \(Simple Merge\)” on page 94](#) can create multiple instances of a task if the incoming paths are not mutually exclusive.

Termination Patterns

Termination patterns are workflow patterns in which the workflow completes normally.

Implicit Termination

The Workflow Service ends processes and subprocesses when all of their activities complete regardless of whether they encounter an End or Terminate end element. For example, suppose you have the workflow shown in [Figure 11.2 on page 89](#). If neither of the conditional expressions defined for the sequence flows on the Start node evaluate to True, the Workflow Service terminates the process.

However, the recommended practice is to explicitly end the paths in your workflow by adding either an End or Terminate end element to each process and subprocess.

Explicit Termination

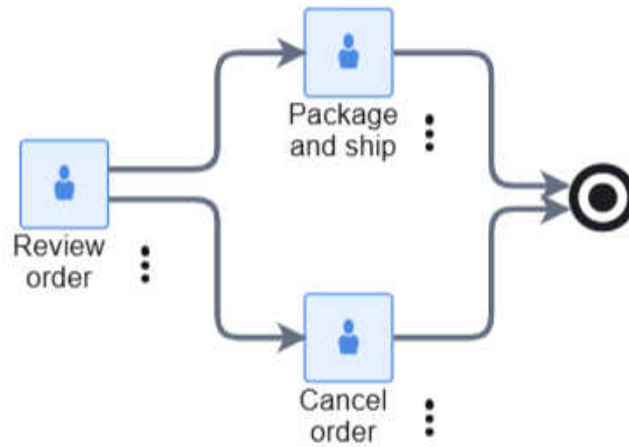
To explicitly end workflow paths, add an End element to each path in the workflow. An End element terminates the path to which it is added, but it does not affect any other paths in workflow.

To explicitly terminate an entire process or subprocess, add a Terminate end element.

IMPORTANT If multiple paths merge into a Terminate end element and the paths are mutually exclusive, a single Terminate end event is triggered. If the paths are not mutually exclusive, a race condition is created. All active paths in the workflow are terminated when the first path reaches the end. The recommended practice is to use an End element. See [“Implicit Termination” on page 98](#).

If you use a Terminate end element, you should either add mutually exclusive conditions to the paths leading to the Terminate End element or merge the paths by using a gateway.

For example, the paths that merge into the Terminate end event in the workflow shown in [Figure 11.10](#) are mutually exclusive.

Figure 11.10 Explicit Termination with Mutually Exclusive Paths

Cancellation Patterns

Cancellation patterns are workflow patterns in which the workflow did not complete normally.

Canceling Activities

You can use a boundary timer, a boundary signal, or a boundary message to cancel the activity with which the timer or message is associated.

For more information, see:

- [“About Timers” on page 65](#)
- [“About Signals” on page 69](#)
- [“About Messages” on page 73](#)

Canceling a Workflow (Cancel Case)

Only workflow administrators can cancel a running instance of a workflow. For more information, see [SAS Workflow Manager: Administrator's Guide](#).

Additional Information about Workflow Patterns

For more information about workflow patterns, see the www.workflowpatterns.com website. The following documents are of particular interest:

- N. Russell, A.H.M. ter Hofstede, W.M.P. van der Aalst, and N. Mulyar. [Workflow Control-Flow Patterns: A Revised View](#). (PDF, 1.04Mb) BPM Center Report BPM-06-22, BPMcenter.org, 2006.
- W.M.P van der Aalst, A.H.M. ter Hofstede, B. Kiepuszewski, and A.P. Barros. [Workflow Patterns](#). (PDF, 718 Kb). Distributed and Parallel Databases, 14(3):5-51, July 2003.

Using the SAS Workflow Manager Copilot

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About SAS Viya Copilot for SAS Workflow Manager

What is SAS Viya Copilot for SAS Workflow Manager?

SAS Viya Copilot for SAS Workflow Manager is intended to assist users in building workflows and managing instances in the application.

The copilot uses an LLM that includes the official SAS documentation for SAS Workflow Manager to help ensure that the responses are relevant and accurate for users. The copilot can provide instructions from the documentation to help you create and understand workflows as a part of business process management. SAS Viya Copilot queries only SAS Workflow Manager documentation and related documentation to find answers. The related documentation that SAS Viya Copilot

for SAS Workflow Manager searches includes the documentation for SAS Model Manager, SAS Intelligent Decisioning, and SAS Clinical Acceleration.

The copilot capabilities in SAS Workflow Manager are surfaced through a conversational chat interface in the **SAS Viya Copilot** pane, as well as directly in the solution itself.

For general information about how SAS Viya Copilot works, as well as information about the guardrails, prompt shields, language detection, and content safety, see [SAS Viya Copilot: Foundation Overview](#).

Enable SAS Viya Copilot Capabilities in SAS Workflow Manager

SAS Viya Copilot is not enabled by default. It must be enabled in SAS Environment Manager by your SAS Viya administrator. When enabled, all of its capabilities are available for your use. For more information about the SAS Viya Copilot page in SAS Environment Manager, see [“SAS Viya Copilot Page” in SAS Environment Manager: User’s Guide](#).

Note: A SAS administrator must set up a credential and secrets before SAS Viya Copilot can be enabled. For more information, see [“Credential and Secrets Setup for SAS Viya Copilot” in SAS Viya Platform: Licensing](#).

Third-Party Models

Model Overview

For a list of the Microsoft Azure models that are used by SAS Viya Copilot, see “Models Used by SAS Viya Copilot” in [SAS Viya Copilot: Foundation Overview](#).

Input and Output Modalities

Input modalities and size limit for each modality as used by SAS:

- gpt-5.1: Text, 272,000 tokens
- gpt-5-nano: Text, 272,000 tokens
- gpt-4.1: Text, 1,047,576 tokens
- gpt-4.1-nano: Text, 1,047,576 tokens

- text-embedding-3-large: Text, 8,192 tokens

Output modalities and size limit for each modality as used by SAS:

- gpt-5.1: Text, 128,000 tokens
- gpt-5-nano: Text, 128,000 tokens
- gpt-4.1: Text, 32,768 tokens
- gpt-4.1-nano: Text, 32,768 tokens
- text-embedding-3-large: Embedded, Fixed-size vector (1536 dimensions)

Note: Because embedding models return a fixed-dimension vector rather than text tokens, there is no published output token size.

Prerequisites

- SAS Workflow Manager 2026.04 or later

Using the Model

- Acceptable use policy: <https://www.sas.com/acceptableuse>.
- Intended use: This feature is a tool to help query the documentation for SAS Workflow Manager.

Type and Nature of AI Systems

The models support features in SAS software, but they cannot be more broadly incorporated into customer AI systems.

Best Practices

You interact with SAS Viya Copilot through the **SAS Viya Copilot** pane. For more information, see [“Getting Assistance from SAS Viya Copilot” on page 104](#).

- Follow these best practices for writing your prompts:
 - Be as clear and specific as possible in the prompts.
 - Be intentional with the tone of your prompts because the AI-generated suggestions often match the tone of the original prompt.
 - Ensure that your prompts are safe and respectful.

- ❑ Enter your prompts only in natural language English.
- ❑ Avoid words or phrases in your prompts that are culturally specific, such as slang or idioms that are not easily understood by others.
- ❑ If you are not getting good results, try rephrasing the prompt.
- ❑ If you need more specific information, ask follow-up questions.

Known Issues or Limitations

- SAS does not guarantee that the AI responses are correct, or that the generated suggestions are optimal.
- LLMs are known to hallucinate.
- SAS Viya Copilot supports only English inputs and output.
- SAS Viya Copilot in SAS Workflow Manager can provide answers only to questions that are related to SAS Workflow Manager. Asking questions about unrelated SAS products or features might produce incorrect responses, or the copilot might state that it is unable to answer the question. See [“What is SAS Viya Copilot for SAS Workflow Manager?”](#) on page 101 for the list of documents that the copilot searches.

Getting Assistance from SAS Viya Copilot

Overview of the SAS Viya Copilot Pane

The SAS Viya Copilot pane provides a conversational interface in which you can use natural English language to interact with SAS Workflow Manager. The SAS Viya Copilot  icon is present in the application bar and when clicked, opens the SAS Viya Copilot pane. You can ask SAS Viya Copilot to retrieve information or instructions from the product documentation. You can click  to close the SAS Viya Copilot pane when finished.

Note: SAS Viya Copilot is not enabled by default. Before you can use this functionality, your SAS administrator must enable it in SAS Environment Manager. For more information, see [“Enable SAS Viya Copilot Capabilities in SAS Workflow Manager”](#) on page 102.

Access the SAS Viya Copilot Pane in SAS Workflow Manager

IMPORTANT English is the only officially supported language for prompts and output. There are no localized versions of SAS Viya Copilot.

To use the copilot:

- 1 Click  on the application bar to open the SAS Viya Copilot pane.
- 2 enter a prompt, or question, and click .

TIP Click  to clear the current conversation and start a new one with the copilot.

Getting the best AI responses can be an iterative process. Your first prompt might not return the best response. If the reply is not what you are looking for, try rewriting your question or asking follow-up questions. Also, because generative AI is non-deterministic, submitting the same question multiple times might generate different answers each time. For more information, see:

- [“Best Practices” on page 103](#)
- [“Best Practices for SAS Viya Copilot” in *SAS Viya Copilot: Foundation Overview*](#)

Example Tasks and Prompts for the SAS Viya Copilot Pane

Get Help with SAS Workflow Manager

You can ask the copilot how to perform tasks in SAS Workflow Manager:

- `How can I see all instances instead of only ones that are in progress?`
- `How can I share a workflow with a specific user?`
- `How do I add tags?`

Manage Your Definitions

You can ask the copilot about different workflow objects and elements:

- `How do I start my workflow?`

.....

Note: Iterations of this example, such as how to run or start a workflow might return information about how workflow instances are typically started in the respective solution that uses the workflow. The Copilot might not return all the information about the prerequisites to starting a workflow instance, such as activating the workflow. In this case, you may need to ask a follow up question, such as `Why can't I start my workflow from SAS Model Manager?` in order to obtain more information.

.....

- `How do I configure a subprocess?`
- `When would I use an exclusive gateway vs. an inclusive gateway?`
- `How do I send a message?`

.....

Note: In this example, the Copilot might return information about [messages on page 73](#), which are an event type that you can use in your definition. However, you might be instead interested in how to send a message, such as a notification or email, to a user associated with a task in your workflow. In that case, consider rephrasing your prompt to `How do I send a message to a user on a task?`.

.....

Manage Instances

You can ask the copilot for information about instances and how to manage them:

- `What permissions do I need to view instances?`
- `How can I progress an instance that is stuck?`
- `What does terminating an instance do?`
- `What are the statuses for a task in an instance?`
- `Why do I keep getting this error?`

.....

Note: The SAS Workflow Manager documentation does not contain much information on error messages. If the copilot is unable to provide assistance related to an error message, you might want to reach out to [SAS Technical Support](#).

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